

Computer Networks

- A computer network is a set of computers sharing resources located on or provided by network nodes. The computers use common communication protocols over digital interconnections to communicate with each other.
- A computer network extends interpersonal communications by electronic means with various technologies, such as email, instant messaging, online chat, voice and video telephone calls, and video conferencing. A network allows sharing of network and computing resources.
- A computer network comprises two or more computers that are connected—either by cables (wired) or WiFi (wireless)—with the purpose of transmitting, exchanging, or sharing data and resources.
- There are different types of computer networks based on the size, scope, and purpose of the network, such as:
 - Local-area network (LAN): A LAN is a collection of connected devices in one physical location, such as a home or an office. A LAN can be small or large, depending on the number of devices and the distance between them. A LAN can be connected to other LANs or to a wide-area network (WAN) through a router.
 - Wide-area network (WAN): A WAN is a collection of connected devices that span a large geographical area, such as a country or a continent. A WAN can be composed of multiple LANs or other smaller networks. A WAN can be connected to other WANs or to the Internet through a gateway.
 - Enterprise network: An enterprise network is a large and complex network that connects multiple LANs and WANs within an organization. An enterprise network can have thousands of devices and multiple layers of switches and routers. An enterprise network can provide various services and applications to the users, such as email, web, database, file sharing, etc.
 - Service-provider network: A service-provider network is a network that provides services to other networks or end users, such as Internet service providers (ISPs), telecommunication companies, cloud providers, etc. A service-provider network can have millions of devices and multiple levels of hierarchy and redundancy. A service-provider network can use various technologies and protocols to deliver the services, such as optical fiber, wireless, satellite, etc.
- Some of the benefits of computer networks are:
 - They enable data and resource sharing among different devices and users, which can improve efficiency and productivity.
 - They facilitate communication and collaboration among different people and groups, which can enhance social and professional interactions.
 - They provide access to a wide range of information and services, which can increase knowledge and opportunities.
 - They support distributed and parallel computing, which can improve

- performance and reliability.
- Some of the challenges of computer networks are:
 - They require proper design, installation, maintenance, and security, which can be costly and complex.
 - They are vulnerable to various threats and attacks, such as malware, hackers, denial-of-service, etc., which can compromise the confidentiality, integrity, and availability of the network and its data.
 - They are subject to various constraints and limitations, such as bandwidth, latency, congestion, etc., which can affect the quality and speed of the network and its services.
- Some of the basic concepts and components of computer networks are:
 - Network topology: The physical or logical arrangement of the devices and links in a network. There are different types of network topologies, such as star, bus, ring, mesh, tree, etc., each with its own advantages and disadvantages.
 - Network protocol: The set of rules and conventions that govern the communication and cooperation among the devices and nodes in a network. There are different types of network protocols, such as TCP/IP, Ethernet, Wi-Fi, HTTP, FTP, etc., each with its own functions and features.
 - Network device: The hardware or software component that performs a specific function or role in a network. There are different types of network devices, such as host, switch, router, gateway, firewall, etc., each with its own capabilities and limitations.
 - Network service: The application or function that provides a specific benefit or functionality to the users or devices in a network. There are different types of network services, such as DNS, DHCP, VPN, NAT, etc., each with its own purpose and mechanism.
- A possible mnemonic to remember the types of network topologies is: **Stars Burn Rapidly Making Tiny Holes** (Star, Bus, Ring, Mesh, Tree, Hybrid).
- A possible mnemonic to remember the layers of the TCP/IP protocol stack is: ****A**

Unit 1 - Introductory Concepts of Computer Networks and Physical Layer

- A computer network is any group of interconnected computing devices capable of sending or receiving data.
- A computing device is not just a computer—it's any device that can run a program, such as a tablet, phone, or smart sensor.
- Computer networks can be classified based on their size, topology, architecture, and protocols.
- Some common types of computer networks are:
 - Local Area Network (LAN): A network that connects devices within a small geographic area, such as a home, office, or school.

- Wide Area Network (WAN): A network that connects devices across a large geographic area, such as a city, country, or the world.
- Metropolitan Area Network (MAN): A network that connects devices within a metropolitan area, such as a city or a campus.
- Personal Area Network (PAN): A network that connects devices within a personal range, such as a Bluetooth headset or a smart-watch.
- Wireless Network: A network that uses wireless technologies, such as radio waves, infrared, or microwaves, to transmit data.
- The Open Systems Interconnection (OSI) model is a conceptual framework that defines the functions of a network system in seven layers.
- The seven layers of the OSI model are:
 - Physical Layer: The lowest layer that deals with the physical and electrical aspects of data transmission, such as cables, connectors, signals, and encoding .
 - Data Link Layer: The layer that provides reliable and error-free data transfer between adjacent nodes, such as switches, bridges, and MAC addresses.
 - Network Layer: The layer that provides logical addressing and routing of data packets across different networks, such as routers, IP addresses, and protocols.
 - Transport Layer: The layer that provides end-to-end data delivery and control, such as TCP, UDP, ports, and flow control.
 - Session Layer: The layer that establishes, maintains, and terminates sessions between applications, such as authentication, synchronization, and dialog control.
 - Presentation Layer: The layer that formats and transforms data for different applications, such as encryption, compression, and character sets.
 - Application Layer: The highest layer that provides user interface and services, such as HTTP, FTP, SMTP, and DNS.
- A mnemonic to remember the seven layers of the OSI model is: **Please Do Not Throw Sausage Pizza Away.**
- The physical layer is responsible for movements of individual bits from one node to the next.
- The physical layer coordinates the functions required to carry a bit stream over a physical medium.
- The physical layer deals with the mechanical and electrical specifications of the interface and the transmission medium.
- Some of the functions of the physical layer are:
 - Bit synchronization: The process of aligning the sender and receiver clocks to ensure correct bit timing.
 - Bit rate: The number of bits transmitted per second.
 - Bandwidth: The range of frequencies available for data transmission.
 - Modulation: The process of changing the characteristics of a carrier signal according to the data signal.

- Multiplexing: The process of combining multiple data streams into one channel.
- Transmission mode: The direction of data flow between two devices, such as simplex, half-duplex, or full-duplex.
- Some of the devices and components used in the physical layer are:
 - Power plugs: The connectors that supply power to the devices.
 - Connectors: The devices that attach the cables to the devices, such as RJ-45, BNC, or USB.
 - Receivers: The devices that receive the signals from the transmission medium, such as antennas, photodetectors, or amplifiers.
 - Cable types: The types of wires or fibers that carry the signals, such as coaxial, twisted pair, or optical fiber.
 - Repeaters: The devices that regenerate and amplify the signals to extend the transmission range.
 - Hubs: The devices that broadcast the signals to all connected devices.

Introductory Concepts of Computer Networks

- A computer network is a system that connects numerous independent computers in order to share information (data) and resources.
- Computer networks help you to connect with multiple computers together to send and receive information.
- Computer networks follow protocols, which define how communications are sent and received.
- Computer networks consist of essential components, such as end devices, media, protocols, and networking devices.
- End devices are devices that send or receive the data in the network, such as computers, printers, and smartphones.
- Media provides connectivity between the end devices, such as cables, fiber optics, or wireless signals .
- Protocols enable communication between two or more end devices, such as TCP/IP, HTTP, and FTP.
- Networking devices are devices that facilitate the transmission and routing of data in the network, such as switches, routers, and wireless access points .
- Computer networks can be classified based on different criteria, such as topology, size, architecture, and functionality.
- Topology refers to the physical or logical arrangement of the network nodes and links, such as bus, ring, star, and mesh.
- Size refers to the geographical area covered by the network, such as local area network (LAN), metropolitan area network (MAN), and wide area network (WAN).
- Architecture refers to the design and structure of the network, such as peer-to-peer (P2P), client-server, and hybrid.
- Functionality refers to the purpose and scope of the network, such as

intranet, extranet, and internet.

Goals and applications of networks and protocols

- A network is a collection of devices that can communicate with each other over a shared medium, such as a cable, a wireless link, or the internet.
- A protocol is a set of rules that defines how devices exchange information over a network, such as the format, timing, and error control of the messages.
- The main goals of networks and protocols are:
 - **Resource sharing:** Networks allow multiple devices to access and use the same resources, such as hardware, software, data, and services. For example, a group of office workers can share a common printer, fax, modem, scanner, etc.
 - **High reliability:** Networks provide multiple sources of supply and backup for the resources, which increases the availability and fault tolerance of the system. For example, if one server fails, another server can take over its tasks.
 - **Greater flexibility:** Networks enable devices to connect and communicate with each other regardless of their physical location, type, or configuration. For example, a laptop can connect to a printer in another room, or a smartphone can access the internet via Wi-Fi or cellular network.
 - **Increased productivity:** Networks facilitate the access and exchange of information among users, which improves the efficiency and collaboration of the work. For example, a team can share files, documents, and messages through a network.
- Some common applications of networks and protocols are:
 - **The internet:** The internet is the largest and most widely used network in the world, which connects billions of devices and users across the globe. The internet uses a suite of protocols called TCP/IP, which defines how data is transmitted, routed, addressed, and secured over the network.
 - **The World Wide Web:** The World Wide Web is a system of interlinked documents and resources that can be accessed through the internet using a web browser. The Web uses a protocol called HTTP, which defines how web servers and web clients communicate and exchange data.
 - **Email:** Email is a system of sending and receiving electronic messages over the internet or other networks. Email uses protocols such as SMTP, POP3, and IMAP, which define how email servers and email clients send, receive, and store messages.

- **File transfer:** File transfer is a system of transferring files or data from one device to another over a network. File transfer uses protocols such as FTP, SFTP, and SCP, which define how files are uploaded, downloaded, and managed over the network.
- **Remote access:** Remote access is a system of accessing or controlling a device or a network from another location over the internet or other networks. Remote access uses protocols such as SSH, Telnet, and RDP, which define how users can log in, execute commands, and view the screen of a remote device.

Categories of networks in computer networks

A computer network is a system of interconnected devices that can communicate and share data, resources, and applications. Computer networks can be classified based on several criteria, such as the size, the topology, the transmission medium, and the organizational intent. Here are some of the common categories of networks in computer networks:

- **Personal Area Network (PAN):** A PAN is the smallest and simplest type of network. It connects devices within the range of an individual, such as a smartphone, a laptop, a printer, or a smartwatch. A PAN is usually no larger than 10 meters and can use wired or wireless technologies, such as Bluetooth, USB, or infrared. A PAN can enable personal communication, data synchronization, and device control. A mnemonic to remember PAN is **P**ersonal **A**ccess to **N**earby devices.
- **Local Area Network (LAN):** A LAN is a network that connects devices within a limited physical area, such as a home, an office, a school, or a laboratory. A LAN can use wired or wireless technologies, such as Ethernet, Wi-Fi, or fiber optics. A LAN can enable fast and reliable data transfer, resource sharing, and network communication. A mnemonic to remember LAN is **L**imited **A**rea **N**etworking.
- **Wide Area Network (WAN):** A WAN is a network that connects devices across a large geographical area, such as a city, a country, or the world. A WAN can use wired or wireless technologies, such as telephone lines, satellite links, or cellular networks. A WAN can enable long-distance communication, data exchange, and internet access. A mnemonic to remember WAN is **W**orldwide **A**rea **N**etworking.
- **Metropolitan Area Network (MAN):** A MAN is a network that connects devices within a metropolitan area, such as a city or a town. A MAN can use wired or wireless technologies, such as cable, DSL, or WiMAX. A MAN can enable high-speed data transfer, internet access, and public services. A mnemonic to remember MAN is **M**etropolitan **A**rea **N**etworking.
- **Campus Area Network (CAN):** A CAN is a network that connects devices within a campus, such as a university, a hospital, or a military base.

A CAN can use wired or wireless technologies, such as Ethernet, Wi-Fi, or fiber optics. A CAN can enable data sharing, network communication, and security. A mnemonic to remember CAN is **C**ampus **A**rea **N**etworking.

- **Storage Area Network (SAN):** A SAN is a network that connects storage devices, such as hard disks, tape drives, or optical disks. A SAN can use wired or wireless technologies, such as SCSI, Fibre Channel, or iSCSI. A SAN can enable high-performance data storage, backup, and recovery. A mnemonic to remember SAN is **S**torage **A**rea **N**etworking.
- **Virtual Private Network (VPN):** A VPN is a network that creates a secure and encrypted connection between devices over a public network, such as the internet. A VPN can use various protocols, such as IPsec, SSL, or PPTP. A VPN can enable remote access, data protection, and anonymity. A mnemonic to remember VPN is **V**irtual **P**rivate **N**etworking.

Organization of the Internet

The Internet is a global network of interconnected computers and devices that communicate using standardized protocols and procedures. The Internet has different levels of organization that affect how it operates and who controls it. These levels are:

- **Computer and network hardware:** This level consists of the physical devices and infrastructure that enable data transmission and routing across the Internet. Examples of hardware include routers, switches, cables, servers, and modems. Hardware can be owned and managed by different entities, such as governments, corporations, universities, or individuals.
- **Internet access and Internet service providers (ISPs):** This level involves the provision of access to the Internet for end users and organizations. ISPs are companies that offer Internet connectivity and related services, such as email, web hosting, and domain name registration. ISPs can vary in size, scope, and quality of service. They can also impose different policies and restrictions on their customers, such as bandwidth limits, content filtering, or censorship.
- **Navigation within the Internet:** This level concerns the methods and tools that help users find and access information and resources on the Internet. Examples of navigation tools include web browsers, search engines, directories, and hyperlinks. Navigation tools can be influenced by various factors, such as algorithms, popularity, relevance, and personalization. They can also be subject to manipulation, bias, or fraud, such as spam, phishing, or fake news.
- **Individual sites and communities:** This level refers to the specific websites, platforms, and online groups that users visit and participate in on the Internet. Examples of sites and communities include social media,

blogs, forums, wikis, and e-commerce. Sites and communities can have their own rules and terms of usage, such as privacy policies, codes of conduct, and moderation systems. They can also create and shape the culture, norms, and values of the Internet users.

ISP

- ISP stands for **Internet Service Provider** , a company that provides access to the internet and other related services to customers.
- ISPs can offer different types of internet connections, such as dial-up, broadband, cable, DSL, fiber-optic, satellite, wireless, etc. Each type has its own advantages and disadvantages in terms of speed, reliability, availability, and cost.
- ISPs can also provide other services, such as email accounts, domain registration, web hosting, browser services, security software, cloud storage, etc. Some ISPs may bundle these services with the internet access or offer them separately.
- ISPs can charge customers based on different factors, such as the type and speed of the internet connection, the amount of data usage, the duration of the contract, the number and quality of the additional services, etc. Some ISPs may offer discounts or incentives for loyal customers or referrals.
- ISPs can host websites for businesses and individuals, and can also build the websites themselves. ISPs can also provide technical support, maintenance, and troubleshooting for the websites and the internet connections.
- ISPs can be classified into different categories, such as Tier 1, Tier 2, and Tier 3 ISPs, based on their network size, coverage, and peering agreements. Tier 1 ISPs are the largest and most interconnected ISPs, which can reach every other network on the internet without paying any fees. Tier 2 ISPs are smaller and less interconnected ISPs, which have to pay fees to Tier 1 ISPs or other Tier 2 ISPs to reach some parts of the internet. Tier 3 ISPs are the smallest and most local ISPs, which have to pay fees to Tier 1 or Tier 2 ISPs to access the internet.
- ISPs can also be classified into different categories, such as access ISPs, hosting ISPs, transit ISPs, virtual ISPs, free ISPs, etc., based on their business model, service offerings, and customer base. Access ISPs provide internet access to customers, hosting ISPs provide web hosting and related services, transit ISPs provide network access to other ISPs, virtual ISPs resell the services of other ISPs, free ISPs provide internet access for free or in exchange for advertising, etc.
- A mnemonic to remember the meaning of ISP is **I**nternet **S**ervice **P**rovider, or **I** Surf the web with **P**rofessional help.

Network structure with reference to Computer Networks

- A computer network is a structure that makes available to a data processing user at one place some data processing function or service performed at another place.
- Computer network structure defines how computers are organized in the network and what tasks are assigned to those computers.
- Computer network structure includes the following components:
 - Hardware: The physical devices that make up the network, such as computers, routers, switches, cables, etc.
 - Software: The programs that run on the hardware and enable communication, such as operating systems, protocols, applications, etc.
 - Transmission media: The medium that carries the data signals between the hardware devices, such as copper wires, optical fibers, radio waves, etc.
 - Network topology: The shape or layout of the network, such as bus, star, ring, mesh, etc.
 - Communications protocols: The rules and standards that govern how data is formatted, transmitted, and received on the network, such as TCP/IP, Ethernet, Wi-Fi, etc.
- Computer network structure can be classified into different types based on the size, scope, and purpose of the network, such as:
 - LAN (local area network): A network that connects computers over a relatively short distance, such as within a building or a campus.
 - WLAN (wireless local area network): A network that uses wireless transmission media, such as radio waves or infrared, to connect computers without cables.
 - MAN (metropolitan area network): A network that covers a larger geographic area, such as a city or a region.
 - WAN (wide area network): A network that spans a very large geographic area, such as a country or a continent.
 - PAN (personal area network): A network that connects personal devices, such as smartphones, tablets, laptops, etc., over a short range, such as within a few meters.
 - SAN (storage area network): A network that connects storage devices, such as hard disks, tapes, etc., to provide high-speed data access and backup.
 - VPN (virtual private network): A network that uses encryption and tunneling techniques to create a secure connection over a public network, such as the Internet.
- Computer network structure can also be classified into different models based on the level of abstraction and functionality of the network, such as:
 - OSI (open systems interconnection) model: A seven-layer model that defines how data is transferred from one computer to another, from the physical layer to the application layer.

- TCP/IP (transmission control protocol/internet protocol) model: A four-layer model that defines how data is transferred over the Internet, from the network access layer to the application layer.
- Hybrid model: A combination of the OSI and TCP/IP models that is commonly used in practice, such as the five-layer model that includes the physical, data link, network, transport, and application layers.
- A mnemonic to remember the seven layers of the OSI model is **Please Do Not Throw Sausage Pizza Away**, where each word corresponds to the first letter of each layer: Physical, Data link, Network, Transport, Session, Presentation, Application.
- A mnemonic to remember the four layers of the TCP/IP model is **NITA**, where each letter corresponds to the first letter of each layer: Network access, Internet, Transport, Application.
- A mnemonic to remember the five layers of the hybrid model is **Please Do Not Touch Anything**, where each word corresponds to the first letter of each layer: Physical, Data link, Network, Transport, Application.

Network architecture with reference to Computer Networks

- Network architecture is the design of a computer network .
- It is a framework for the specification of a network's physical components and their functional organization and configuration, its operational principles and procedures, as well as communication protocols used .
- Network architecture components include hardware, software, transmission media (wired or wireless), network topology, and communications protocols .
- Based on the network's size and purpose, numerous ways to design network architecture exist .
- Here are the most common and widely used computer network types :
 - LAN (local area network): A LAN connects computers over a relatively short distance, allowing them to share data, files, printers, and other resources. A LAN typically covers a single building or a group of buildings in a campus. A LAN can be wired or wireless. An example of a LAN is a home network or a school network.
 - WLAN (wireless local area network): A WLAN is just like a LAN but connections between devices on the network are wireless, using radio waves or infrared signals. A WLAN can provide mobility and flexibility to the users. A WLAN typically uses a wireless access point (AP) to connect to a wired network and provide wireless coverage to a certain area. An example of a WLAN is a Wi-Fi network or a hotspot.
 - MAN (metropolitan area network): A MAN connects computers over a larger geographic area than a LAN, such as a city or a town. A MAN can span several kilometers and can interconnect several LANs. A MAN can use various technologies, such as fiber-optic cables, microwave links, or satellite links. An example of a MAN is a cable TV

network or a municipal network.

- WAN (wide area network): A WAN connects computers over a very large geographic area, such as a country or a continent. A WAN can span thousands of kilometers and can interconnect several LANs and MANs. A WAN can use various technologies, such as leased lines, public switched telephone network (PSTN), cellular networks, or the Internet. An example of a WAN is the Internet or a corporate network.
- PAN (personal area network): A PAN connects devices over a very short distance, such as a few meters. A PAN can provide personal connectivity to the user and can support various devices, such as laptops, smartphones, tablets, printers, scanners, or wearable devices. A PAN can be wired or wireless. An example of a PAN is a Bluetooth network or a USB network.
- SAN (storage area network): A SAN connects storage devices, such as hard disks, tape drives, or optical drives, to servers or other computers. A SAN can provide high-speed data transfer, high availability, and scalability to the storage devices. A SAN can use various technologies, such as fiber channel, iSCSI, or SCSI. An example of a SAN is a data center network or a backup network.
- CAN (campus area network): A CAN connects computers and devices within a limited geographic area, such as a university campus or a military base. A CAN can be considered as a larger LAN or a smaller MAN. A CAN can use various technologies, such as Ethernet, Wi-Fi, or ATM. An example of a CAN is a university network or a military network.
- VPN (virtual private network): A VPN connects computers and devices over a public network, such as the Internet, in a secure and private way. A VPN can provide encryption, authentication, and tunneling to the data traffic. A VPN can use various technologies, such as IPsec, SSL, or PPTP. An example of a VPN is a remote access network or a site-to-site network.
- A mnemonic to remember the network types is **LWMWPSCV** (pronounced as “Lum-Wum-Pus-Cee-Vee”). The first letter of each network type forms the mnemonic. For example, L stands for LAN, W stands for WLAN, M stands for MAN, and so on.

Layering principles with reference to Network Architecture in Computer Networks

- Layering is a mechanism that divides a complex system into smaller and manageable parts, called layers, that interact with each other through well-defined interfaces.
- Layering helps to achieve modularity, abstraction, encapsulation, and interoperability in network design and implementation.
- Layering also facilitates the development and evolution of network proto-

cols and standards, as each layer can be modified or replaced independently of the others, as long as the interfaces are preserved.

- There are different models of network architecture that use layering, such as the Open Systems Interconnection (OSI) model and the Transmission Control Protocol/Internet Protocol (TCP/IP) model.
- The OSI model defines seven layers of network functions, from the physical layer that deals with the transmission of bits over a medium, to the application layer that provides services to end-user applications.
- The TCP/IP model is a simpler and more widely used model that consists of four layers: the network access layer, the internet layer, the transport layer, and the application layer.
- Each layer in a network model has a specific role and responsibility, and provides a service to the layer above it and uses the service of the layer below it.
- For example, the transport layer ensures reliable and ordered delivery of data packets between two end hosts, and uses the internet layer to route the packets across different networks.
- The layering principles can be summarized as follows:
 - A layer should be created where a different abstraction is needed.
 - Each layer should perform a well-defined function.
 - The function of each layer should be chosen with an eye toward defining internationally standardized protocols.
 - The layer boundaries should be chosen to minimize the information flow across the interfaces.
 - The number of layers should be large enough that distinct functions need not be thrown together in the same layer out of necessity, and small enough that the architecture does not become unwieldy.
- A possible mnemonic to remember the names of the OSI layers is **All People Seem To Need Data Processing**, where the first letter of each word corresponds to the first letter of each layer, from top to bottom: Application, Presentation, Session, Transport, Network, Data Link, Physical.
- A possible mnemonic to remember the names of the TCP/IP layers is **A Thin Ice Needs Care**, where the first letter of each word corresponds to the first letter of each layer, from top to bottom: Application, Transport, Internet, Network Access.

Services in networks architecture in Computer Networks

- Services in networks architecture are the **functions** or **operations** that each layer of the network model provides to the layer above it or below it.
- Services are **abstract** and **independent** of the implementation details of the layer that provides them or the layer that uses them.
- Services are **defined** by a set of **primitives** or **commands** that specify the **type**, **parameters**, and **results** of the service.
- Services are **invoked** by the **service user** (the layer above) and **per-**

formed by the **service provider** (the layer below) through the **service access point** (SAP), which is a logical point of interaction between the layers.

- Services can be **classified** into two types: **connection-oriented** and **connectionless**.
 - Connection-oriented services require the establishment of a **logical connection** between the service user and the service provider before any data can be exchanged. The connection is **terminated** after the data transfer is complete. Connection-oriented services provide **reliable, ordered, and error-free** delivery of data, but they also incur **overhead** and **delay** for connection setup and teardown.
 - Connectionless services do not require any connection establishment or termination. The service user can send data to the service provider at any time, without any prior agreement. Connectionless services provide **fast** and **efficient** delivery of data, but they do not guarantee **reliability, order, or error-correction**.
- Examples of services in networks architecture are:
 - **DHCP** (Dynamic Host Configuration Protocol): A service that assigns IP addresses and other network configuration parameters to hosts dynamically.
 - **DNS** (Domain Name System): A service that maps domain names to IP addresses and vice versa.
 - **FTP** (File Transfer Protocol): A service that allows users to transfer files between hosts over a network.
 - **HTTP** (Hypertext Transfer Protocol): A service that allows users to access web pages and other resources over a network.
 - **SMTP** (Simple Mail Transfer Protocol): A service that allows users to send and receive email messages over a network.

Protocols and standards in networks architecture in Computer Networks

- Network protocols are the rules and procedures that define how devices communicate and exchange data over a network. They specify the format, timing, sequencing, and error control of the data packets that are transmitted and received by the devices.
- Network standards are the agreed-upon conventions and guidelines that ensure interoperability and compatibility among different devices, vendors, and technologies in a network. They are developed and maintained by various organizations and committees, such as IEEE, IETF, ISO, etc.
- Network architecture is the design and structure of a network, including its hardware, software, transmission media, topology, and protocols. It determines how the network components interact and cooperate to achieve the network's goals and functions.

- Some of the common network protocols and standards are:
 - TCP/IP: Transmission Control Protocol/Internet Protocol is the most widely used protocol suite in the Internet and other networks. It consists of four layers: application, transport, internet, and network access. It provides reliable, connection-oriented, and end-to-end data delivery, as well as addressing, routing, and fragmentation functions.
 - UDP: User Datagram Protocol is a connectionless and unreliable transport protocol that offers low-overhead and fast data transmission. It is suitable for real-time applications, such as voice and video streaming, that can tolerate some data loss and delay.
 - HTTP: Hypertext Transfer Protocol is an application layer protocol that defines how web browsers and web servers communicate and exchange web resources, such as HTML documents, images, and scripts. It uses a request-response model, where the browser sends an HTTP request to the server, and the server responds with an HTTP response containing the requested resource or an error message.
 - FTP: File Transfer Protocol is an application layer protocol that enables the transfer of files between a client and a server over a TCP connection. It supports both binary and ASCII modes, as well as authentication, authorization, and directory listing functions.
 - DNS: Domain Name System is a distributed and hierarchical system that maps domain names, such as `www.google.com`, to their corresponding IP addresses, such as `172.217.14.206`. It uses a network of name servers that store and resolve the domain name queries from the clients.
 - DHCP: Dynamic Host Configuration Protocol is an application layer protocol that assigns IP addresses and other network configuration parameters to the devices in a network dynamically. It uses a four-step process: discover, offer, request, and acknowledge, to allocate and renew the IP leases for the devices.
 - ARP: Address Resolution Protocol is a network access layer protocol that resolves the IP addresses to their corresponding MAC addresses, which are the physical addresses of the network interface cards. It uses a two-step process: request and reply, to broadcast and cache the ARP entries for the devices in a local network.
 - IEEE 802: This is a family of standards that define the physical and data link layer specifications for various types of networks, such as Ethernet, Wi-Fi, Bluetooth, etc. They specify the characteristics of the transmission media, the encoding and modulation schemes, the frame formats, the error detection and correction methods, the access control and flow control mechanisms, etc.

The OSI reference model in in Computer Networks

- The OSI (Open Systems Interconnection) model is a reference model for how applications communicate over a network.
- It was developed by the International Organization for Standardization (ISO) in 1980.
- It consists of seven layers, each of which performs a specific function in the communication process.
- The layers are, from bottom to top:
 - Layer 1: Physical layer
 - * This layer defines the physical characteristics of the transmission medium, such as voltage levels, connectors, cables, etc.
 - * It also deals with the transmission and reception of raw bits over the medium.
 - * Examples of devices that operate at this layer are hubs, repeaters, and network interface cards (NICs).
 - Layer 2: Data link layer
 - * This layer provides reliable and error-free transmission of data frames between adjacent nodes on the same network segment.
 - * It also handles the addressing, framing, and flow control of data packets.
 - * Examples of devices that operate at this layer are switches, bridges, and network interface cards (NICs).
 - Layer 3: Network layer
 - * This layer provides logical addressing and routing of data packets across different networks or subnets.
 - * It also handles the fragmentation and reassembly of packets, as well as the congestion control and quality of service (QoS) of the network.
 - * Examples of devices that operate at this layer are routers, firewalls, and network interface cards (NICs).
 - Layer 4: Transport layer
 - * This layer provides end-to-end communication between applications on different hosts.
 - * It also handles the segmentation and reassembly of data streams, as well as the error detection and correction, and flow control of the communication.
 - * Examples of protocols that operate at this layer are TCP (Transmission Control Protocol) and UDP (User Datagram Protocol).
 - Layer 5: Session layer
 - * This layer establishes, maintains, and terminates sessions between applications on different hosts.
 - * It also handles the synchronization, authentication, and authorization of the communication.
 - * Examples of protocols that operate at this layer are RPC (Remote Procedure Call) and NFS (Network File System).
 - Layer 6: Presentation layer
 - * This layer provides the translation, encryption, and compression

- of data between different formats and standards.
 - * It also handles the representation and interpretation of data for the application layer.
 - * Examples of protocols that operate at this layer are SSL (Secure Sockets Layer) and MIME (Multipurpose Internet Mail Extensions).
- Layer 7: Application layer
 - * This layer provides the interface and services for the user applications to access the network resources.
 - * It also handles the high-level functions and protocols of the communication.
 - * Examples of protocols that operate at this layer are HTTP (Hypertext Transfer Protocol), FTP (File Transfer Protocol), and SMTP (Simple Mail Transfer Protocol).
- The OSI model is useful for understanding the functions and interactions of different network components and protocols.
- It also provides a common framework and terminology for network design and troubleshooting.
- However, the OSI model is not a strict standard that all network devices and protocols must follow. Some protocols may span across multiple layers, or omit some layers altogether.
- A common mnemonic to remember the names and order of the OSI layers is “Please Do Not Throw Sausage Pizza Away”, where the first letter of each word corresponds to the first letter of each layer.

TCP/IP protocol suite in computer networks

- TCP/IP stands for Transmission Control Protocol/Internet Protocol and is a suite of communication protocols used to interconnect network devices on the internet or a private network .
- TCP/IP is also known as the internet protocol suite, as it defines the standards and rules for data transmission over the internet .
- TCP/IP consists of four layers: application, transport, internet, and network interface .
- The application layer provides the interface for the user or the application to access the network services, such as email, web browsing, file transfer, etc. Some of the common protocols in this layer are HTTP, SMTP, FTP, DNS, etc .
- The transport layer provides the end-to-end data delivery between the source and the destination hosts, by ensuring reliable, ordered, and error-free transmission. The two main protocols in this layer are TCP and UDP .
- TCP (Transmission Control Protocol) is a connection-oriented protocol

that establishes a logical connection between the sender and the receiver, and uses sequence numbers, acknowledgements, and retransmissions to ensure reliable data transfer .

- UDP (User Datagram Protocol) is a connectionless protocol that does not guarantee reliable data delivery, but is faster and more efficient than TCP. UDP is suitable for real-time applications, such as video streaming, voice over IP, etc .
- The internet layer is responsible for the routing and forwarding of packets across different networks, by using the IP (Internet Protocol) address of the source and the destination hosts. The IP address is a unique identifier for each host on the internet, and is usually written in dotted decimal notation, such as 192.168.1.1 .
- The IP (Internet Protocol) is the core protocol of the internet layer, and defines the format and structure of the packets, as well as the rules for addressing and fragmenting them. The current version of IP is IPv4, which uses 32-bit addresses, but a newer version, IPv6, which uses 128-bit addresses, is being deployed to overcome the address space limitation .
- The network interface layer is the lowest layer of the TCP/IP suite, and interacts with the physical network devices, such as the network interface card (NIC), the cable, the hub, the switch, etc. This layer is also known as the link layer or the data link layer, and handles the framing, encoding, and error detection of the data bits .
- The network interface layer can use different protocols depending on the type and topology of the network, such as Ethernet, Wi-Fi, Bluetooth, etc .
- A mnemonic to remember the four layers of the TCP/IP suite is **A Tea In November**, where A stands for Application, T stands for Transport, I stands for Internet, and N stands for Network interface.
- A diagram to illustrate the TCP/IP suite and its protocols is shown below:

Application					Transport		Internet				Network Interface
HTTP	SMTP	FTP	DNS	TCP	UDP		IP	ICMP	ARP	RARP	Ethernet

Network devices in Computer Networks

Network devices are physical devices that enable communication and interaction between hardware on a computer network. Each networking device operates in a distinct computer network segment and performs distinct functions. A network may require hundreds or thousands of different network devices to maintain and build out various LAN and WAN.

Some of the common types of network devices are:

- **Repeater:** A repeater is a device that operates at the physical layer and regenerates the signal over the same network. It can extend the transmission distance of a network by amplifying the weak signals. However, it cannot filter the traffic or prevent collisions.
- **Hub:** A hub is a device that operates at the physical layer and connects multiple wires coming from different branches. It acts as a multiport repeater and broadcasts the incoming data to all the ports. It is simple and cheap, but it creates a lot of noise and congestion in the network.
- **Bridge:** A bridge is a device that operates at the data link layer and connects two or more network segments. It can filter the traffic based on the MAC addresses and forward the data to the appropriate segment. It can also reduce collisions and increase the network performance.
- **Switch:** A switch is a device that operates at the data link layer and connects multiple devices in a network. It can store the MAC addresses of the connected devices in a table and forward the data to the specific port. It can also create separate collision domains and improve the network efficiency.
- **Router:** A router is a device that operates at the network layer and routes the data packets based on their IP addresses. It can connect different networks with different protocols and architectures. It can also perform network address translation (NAT), firewall, and encryption functions.
- **Gateway:** A gateway is a device that operates at the application layer and converts the data from one format to another. It can connect different networks with different protocols and applications. It can also perform protocol conversion, data encryption, and authentication functions.
- **Brouter:** A brouter is a device that combines the functions of a bridge and a router. It can filter the traffic at the data link layer and route the traffic at the network layer. It can also create separate broadcast domains and reduce the network congestion.
- **NIC:** A network interface card (NIC) is a device that operates at the physical and data link layers and connects a computer to a network. It can convert the digital signals from the computer to the electrical signals for the network and vice versa. It can also provide the MAC address for the computer and perform error detection and correction functions.

Some of the mnemonics and learning tricks for the network devices are:

- To remember the order of the network devices from the lowest to the highest layer, use the phrase **RHBS RGBN** (Repeater, Hub, Bridge, Switch, Router, Gateway, Brouter, NIC).
- To remember the functions of the network devices, use the acronym **RAF-FLE** (Regenerate, Amplify, Filter, Forward, Link, Encrypt).
- To remember the difference between a hub and a switch, use the word **BUSH** (Broadcast, Unicast, Switch, Hub). A hub broadcasts the data to all the ports, while a switch unicasts the data to the specific port.

Network components in Computer Networks

Network components are the devices and software that enable communication and data exchange between computers and other networked devices. Some of the major network components are:

- **Network Interface Card (NIC):** A NIC is a hardware device that allows a computer to connect to a network. It provides a physical interface for sending and receiving data over a network medium, such as a cable or wireless signal. A NIC has a unique address, called a MAC address, that identifies the device on the network. A NIC can also perform some data processing functions, such as error detection and correction, encryption and decryption, and packet filtering.
- **Hub:** A hub is a device that connects multiple computers or other network devices together. It acts as a central point for data transmission, and broadcasts any data it receives to all the connected devices. A hub operates at the physical layer of the network, and does not perform any data processing or filtering. A hub is a simple and cheap device, but it can cause network congestion and collisions, as it does not distinguish between different destinations or sources of data.
- **Switch:** A switch is a device that connects multiple computers or other network devices together. It acts as a central point for data transmission, but unlike a hub, it can forward data to specific destinations based on the MAC addresses of the devices. A switch operates at the data link layer of the network, and can perform some data processing and filtering functions, such as VLANs, QoS, and security. A switch is a more intelligent and efficient device than a hub, as it can reduce network congestion and collisions, and improve network performance and security.
- **Cables and connectors:** Cables and connectors are the physical components that link the network devices together. They provide the medium for data transmission, and can be of different types, such as twisted pair, coaxial, fiber optic, or wireless. Cables and connectors have different characteristics, such as bandwidth, latency, noise, interference, and distance limitations, that affect the quality and speed of data transmission.
- **Router:** A router is a device that connects multiple networks together. It acts as a gateway for data transmission, and can forward data to different destinations based on the IP addresses of the devices. A router operates at the network layer of the network, and can perform some data processing and routing functions, such as NAT, DHCP, firewall, and VPN. A router is a more intelligent and complex device than a switch, as it can handle different network protocols, topologies, and architectures, and improve network connectivity and security.
- **Modem:** A modem is a device that modulates and demodulates analog signals into digital signals, and vice versa. It enables data transmission

over different types of media, such as telephone lines, cable lines, or satellite signals. A modem operates at the physical layer of the network, and can perform some data processing and conversion functions, such as compression, encryption, and error correction. A modem is a necessary device for connecting to the Internet, as it can convert the digital signals from the computer to the analog signals from the Internet service provider.

- **Server:** A server is a computer that provides services or resources to other computers or devices on the network. It acts as a central point for data storage, processing, and distribution, and can perform various functions, such as web hosting, email, file sharing, database, printing, and backup. A server operates at the application layer of the network, and can run different software and protocols, such as HTTP, SMTP, FTP, SQL, and DNS. A server is a powerful and specialized device that can handle multiple requests and tasks from the network clients.
- **Client:** A client is a computer or device that requests services or resources from a server on the network. It acts as an end point for data transmission, and can perform various functions, such as browsing, emailing, downloading, uploading, and printing. A client operates at the application layer of the network, and can run different software and protocols, such as web browsers, email clients, file transfer clients, and print drivers. A client is a general-purpose and user-friendly device that can access and manipulate the data from the network servers.
- **Firewall:** A firewall is a device

Physical Layer in Computer Networks

- The physical layer is the first and lowest layer of the OSI model of computer networking.
- The physical layer provides an electrical, mechanical, and procedural interface to the transmission medium.
- The physical layer defines the means of transmitting a stream of raw bits over a physical data link connecting network nodes.
- The physical layer consists of the electronic circuit transmission technologies of a network.
- The physical layer maintains the data rate (how many bits a sender can send per second).
- The physical layer performs synchronization of bits.
- The physical layer helps in transmission medium decision (direction of data transfer).
- The physical layer helps in physical topology (mesh, star, bus, ring) decision (topology through which we can connect the devices).
- The physical layer is responsible for encoding, modulation, multiplexing, signal generation, transmission, reception, and demodulation of signals.
- The physical layer can be implemented through a great number of different

hardware technologies with widely varying characteristics.

- Some examples of physical layer devices are hubs, repeaters, cables, connectors, and network interface cards.
- Some examples of physical layer standards are Ethernet, Wi-Fi, Bluetooth, USB, and RS-232.

Mnemonics and learning tricks for the physical layer

- To remember the functions of the physical layer, you can use the acronym **EMT DR SED**:
 - **E**ncoding
 - **M**odulation
 - **T**ransmission
 - **D**ata rate
 - **R**eception
 - **S**ynchronization
 - **E**lectrical, mechanical, and procedural interface
 - **D**emodulation
- To remember the order of the OSI model layers, you can use the mnemonic **Please Do Not Throw Sausage Pizza Away**:
 - **P**hysical
 - **D**ata Link
 - **N**etwork
 - **T**ransport
 - **S**ession
 - **P**resentation
 - **A**pplication
- To remember some examples of physical layer devices, you can use the mnemonic **HRC CN**:
 - **H**ubs
 - **R**epeaters
 - **C**ables
 - **C**onnectors
 - **N**etwork interface cards
- To remember some examples of physical layer standards, you can use the mnemonic **EW BUS**:
 - **E**thernet
 - **W**i-Fi
 - **B**luetooth
 - **U**SB
 - **S**-232

Network topology design in Computer Networks

- Network topology is the physical and logical arrangement of network devices and how they communicate with each other .
- Network topology affects the performance, cost, reliability, security, and scalability of a network.
- There are two main types of network topology: physical and logical.
 - Physical topology refers to the actual layout of the network devices and cables, such as bus, ring, star, mesh, tree, etc.
 - Logical topology refers to the way data flows through the network, regardless of the physical layout, such as broadcast, multicast, unicast, etc.
- The choice of network topology depends on various factors, such as:
 - The size and scope of the network.
 - The bandwidth and latency requirements.
 - The budget and resources available.
 - The security and redundancy needs.
 - The ease of installation and maintenance.
- Some common network topologies and their characteristics are:

Topology	Description	Advantages	Disadvantages	Example
Bus	All devices are connected to a single cable called the bus or backbone.	Simple and cheap to install and expand.	Low bandwidth and high collision rate. Difficult to troubleshoot and isolate faults. If the bus breaks, the whole network fails.	Ethernet networks.
Ring	All devices are connected to a single closed loop of cable. Data travels in one direction around the ring.	High bandwidth and low collision rate. Easy to troubleshoot and isolate faults.	More expensive and complex to install and expand than bus. If one device or cable fails, the whole network fails.	Token Ring networks.

Topology	Description	Advantages	Disadvantages	Example
Star	All devices are connected to a central device called the hub or switch. Data travels from one device to the hub and then to another device.	High bandwidth and low collision rate. Easy to install, expand, and modify. Faults can be easily detected and isolated.	More expensive and requires more cable than bus or ring. If the hub fails, the whole network fails.	Most LANs today use star topology.
Mesh	All devices are connected to each other by multiple paths. Data can travel along the shortest and least congested path.	High bandwidth and reliability. Faults can be easily detected and bypassed.	Very expensive and complex to install and maintain. Requires a lot of cable and ports.	WANs and MANs use mesh topology.
Tree	A hierarchical structure of star networks connected to a bus network.	Combines the advantages of bus and star topologies. Easy to expand and manage.	More expensive and complex than bus or star topologies. If the bus or the root hub fails, the whole network fails.	Cable TV networks use tree topology.

- A mnemonic to remember the five common network topologies is: **Busy Roads Start Making Trouble** (Bus, Ring, Star, Mesh, Tree).
- A network topology diagram is a graphical representation of the physical and logical layout of a network. It helps to visualize the network structure, identify the devices and links, and troubleshoot the network problems.
- A network topology diagram can be created using various tools, such as Microsoft Visio, Lucidchart, Draw.io, etc.
- A network topology diagram should include the following elements:
 - The network devices, such as routers, switches, hubs, firewalls, servers, workstations, etc.
 - The network cables, such as coaxial, twisted pair, fiber optic, etc.
 - The network symbols, such as circles, squares, triangles, etc.
 - The network labels, such as device names, IP addresses, port numbers, etc.

- The network legend, such as color codes

Types of connections in computer networks

- A connection in a computer network is a link between two or more devices that allows them to communicate and share data, resources, and applications.
- There are different types of connections in computer networks, depending on the size, scope, and function of the network. Some of the common types are:
 - **Point-to-point connection:** This type of connection allows one device to communicate with one other device. For example, two phones may pair with each other using Bluetooth to exchange files or make calls. A point-to-point connection is usually established by a cable or a wireless signal. A point-to-point connection is simple, secure, and efficient, but it does not scale well for large networks.
 - **Broadcast/multicast connection:** This type of connection allows a device to send one message out to the network and have copies of that message delivered to multiple recipients. For example, a radio station may broadcast its program to many listeners, or a video conference may multicast its stream to many participants. A broadcast/multicast connection is usually established by a wireless signal or a shared medium, such as a coaxial cable. A broadcast/multicast connection is useful for disseminating information to a large group, but it may cause congestion and interference in the network.
 - **Multipoint connection:** This type of connection allows one device to connect and deliver messages to multiple devices in parallel. For example, a router may connect to several computers and forward packets to them based on their destination addresses. A multipoint connection is usually established by a switch or a hub, which are devices that can split and join signals. A multipoint connection is flexible and scalable, but it may require more hardware and management.
- A mnemonic to remember the types of connections in computer networks is **PBM** (Point-to-point, Broadcast/multicast, Multipoint). You can also think of them as **one-to-one, one-to-many, and many-to-many** connections.

Transmission media in Computer Networks

Transmission media in computer networks are the physical channels that connect the sender and the receiver of data. They are used to transfer data in the form of bits or signals over a network. Transmission media can be classified into two main types: guided media and unguided media .

- Guided media: These are the media that provide a physical path for the data to travel along. They are also called wired or bounded media. Examples of guided media are twisted pair cables, coaxial cables, and optical fibers. Guided media have the advantages of high bandwidth, low attenuation, and low interference. However, they also have the disadvantages of high cost, limited length, and installation complexity .
- Unguided media: These are the media that do not provide a physical path for the data to travel along. They are also called wireless or unbounded media. Examples of unguided media are radio waves, microwaves, infrared waves, and visible light. Unguided media have the advantages of low cost, easy installation, and mobility. However, they also have the disadvantages of low bandwidth, high attenuation, and high interference .

Transmission media are an essential component of network architecture, which is the design and structure of a network. Network architecture includes hardware, software, protocols, and topology . The choice of transmission media depends on various factors, such as the distance, speed, reliability, security, and cost of data transmission. Different transmission media have different characteristics and suitability for different network applications .

Signal transmission and encoding in Computer Networks

- Signal transmission is the process of sending and receiving data over a physical medium, such as a wire, a fiber optic cable, or a wireless channel.
- Encoding is the process of converting data into a format that can be transmitted as a signal, such as a voltage level, a light pulse, or a radio wave.
- Encoding can be classified into two types: analog-to-digital encoding and digital-to-digital encoding.
- Analog-to-digital encoding is the process of converting an analog signal, such as a voice or a video, into a digital signal, such as a sequence of bits. This involves three steps: sampling, quantization, and encoding.
 - Sampling is the process of measuring the amplitude of the analog signal at regular intervals, called the sampling rate. The sampling rate should be at least twice the highest frequency of the analog signal, according to the Nyquist theorem, to avoid aliasing.
 - Quantization is the process of assigning a discrete value, called a quantization level, to each sample. The quantization level can be represented by a fixed number of bits, called the bit depth. The bit depth determines the resolution and the dynamic range of the digital signal. The quantization error is the difference between the actual sample value and the quantization level.
 - Encoding is the process of converting each quantization level into a binary code, such as a pulse code modulation (PCM) code. The encoding scheme can be uniform or non-uniform, depending on whether the quantization levels are equally or unequally spaced. Non-uniform

encoding schemes, such as mu-law and A-law, can reduce the quantization error for low-amplitude signals.

- Digital-to-digital encoding is the process of converting a digital signal, such as a sequence of bits, into another digital signal, such as a series of voltage pulses. This is also known as line coding. The purpose of line coding is to facilitate the transmission, detection, and synchronization of the digital signal. Some common line coding schemes are:
 - Unipolar encoding: The digital signal is represented by two voltage levels, one for 0 and one for 1. For example, non-return-to-zero (NRZ) encoding uses a high voltage for 1 and a low voltage for 0. The advantage of unipolar encoding is its simplicity and low power consumption. The disadvantage is its lack of synchronization and its susceptibility to noise and baseline wander.
 - Bipolar encoding: The digital signal is represented by three voltage levels, one for 0 and two for 1. For example, alternate mark inversion (AMI) encoding uses a high voltage for 1, a low voltage for 0, and alternates the polarity of the voltage for consecutive 1s. The advantage of bipolar encoding is its better synchronization and noise immunity. The disadvantage is its higher power consumption and its problem of long sequences of 0s.
 - Manchester encoding: The digital signal is represented by a transition in the voltage level at the middle of each bit period. For example, a high-to-low transition represents 1, and a low-to-high transition represents 0. The advantage of Manchester encoding is its inherent synchronization and its self-clocking property. The disadvantage is its double bandwidth requirement and its complexity.
- Some mnemonics and learning tricks for signal transmission and encoding in computer networks are:
 - To remember the Nyquist theorem, think of “twice as nice” as the sampling rate should be twice the highest frequency of the analog signal.
 - To remember the difference between PCM and AMI, think of “PCM is plain, AMI is alternating” as PCM uses a uniform encoding scheme, while AMI uses a non-uniform encoding scheme that alternates the polarity of the voltage for consecutive 1s.
 - To remember the difference between NRZ and Manchester encoding, think of “NRZ is no return, Manchester is mid transition” as NRZ encoding does not return to the zero voltage level, while Manchester encoding has a transition at the middle of each bit period.

Network performance and transmission impairments in Computer Networks

- Network performance is the measure of how well a network can deliver data and services to its users. It can be affected by various factors, such as bandwidth, latency, jitter, packet loss, and error rate.

- Transmission impairments are the factors that degrade the quality and reliability of a signal as it travels through a transmission medium. They can cause distortion, attenuation, or noise in the signal, which may result in errors or loss of data.
- There are three main types of transmission impairments: attenuation, delay distortion, and noise.
 - Attenuation is the loss of signal strength as it travels over a distance. It can be caused by the resistance of the medium, the absorption of the signal by the medium, or the leakage of the signal from the medium. Attenuation can be measured in decibels (dB), which is the ratio of the input power to the output power of the signal. Attenuation can be compensated by using amplifiers or repeaters to boost the signal strength.
 - Delay distortion is the variation of signal delay as it travels through a medium. It can be caused by the different propagation speeds of different frequency components of the signal, or by the interference of multiple signals in the same medium. Delay distortion can affect the synchronization and alignment of the signal, which may result in errors or loss of data. Delay distortion can be reduced by using equalizers or filters to balance the signal delay.
 - Noise is the unwanted or random variation of the signal due to external or internal sources. It can be classified into three types: thermal noise, crosstalk, and impulse noise.
 - * Thermal noise is the random fluctuation of the signal due to the thermal agitation of electrons in the medium. It is proportional to the temperature and the bandwidth of the signal, and it cannot be eliminated completely. Thermal noise can be minimized by using a high signal-to-noise ratio (SNR), which is the ratio of the signal power to the noise power.
 - * Crosstalk is the interference of one signal by another signal in the same or adjacent medium. It can be caused by the coupling of electromagnetic fields or the leakage of signals from one medium to another. Crosstalk can be measured in dB, which is the ratio of the unwanted signal power to the desired signal power. Crosstalk can be avoided by using proper shielding, isolation, or modulation techniques.
 - * Impulse noise is the sudden and brief disturbance of the signal due to external sources, such as lightning, switching, or sparks. It can cause spikes or bursts of noise in the signal, which may result in errors or loss of data. Impulse noise can be detected and corrected by using error detection and correction techniques, such as parity bits, checksums, or codes.

Switching techniques and multiplexing in Computer Networks

Switching techniques are methods of connecting multiple devices in a network

and transferring data between them. Multiplexing is a technique of combining multiple signals into one signal over a shared medium. Both techniques are used to improve the efficiency and performance of communication networks.

There are three main types of switching techniques:

- **Circuit switching:** In circuit switching, two nodes communicate with each other over a dedicated communication path. The path is established before the data transmission and remains active until the communication is over. The path is reserved exclusively for the two nodes and no other node can use it. Circuit switching is reliable and guarantees a constant bandwidth, but it is inefficient and costly. An example of circuit switching is the traditional telephone network.
- **Message switching:** In message switching, the whole message is treated as a data unit. The message is stored and forwarded by intermediate nodes until it reaches the destination. The message may take different paths and may arrive out of order. There is no dedicated path or reservation of bandwidth. Message switching is flexible and can handle variable-length messages, but it introduces delay and overhead. An example of message switching is email.
- **Packet switching:** The packet switching technique is derived from message switching where the message is broken down into smaller chunks called packets. Each packet has a header that contains the source and destination addresses and other information. The packets are routed independently by intermediate nodes and may take different paths. The packets are reassembled at the destination. There is no dedicated path or reservation of bandwidth. Packet switching is efficient and scalable, but it may cause congestion and loss of packets. An example of packet switching is the internet.

Multiplexing can be done in different ways, depending on the type of signal and the medium. Some common types of multiplexing are:

- **Frequency division multiplexing (FDM):** In FDM, the frequency spectrum of the medium is divided into several non-overlapping frequency bands. Each signal is modulated by a different carrier frequency and transmitted over a different band. The signals are separated at the receiver by using filters. FDM is suitable for analog signals and can be used in radio, television, and cable networks.
- **Time division multiplexing (TDM):** In TDM, the time axis of the medium is divided into several time slots. Each signal is assigned a different time slot and transmitted over the medium in a round-robin fashion. The signals are separated at the receiver by using synchronization. TDM is suitable for digital signals and can be used in telephone, cellular, and optical networks.
- **Statistical multiplexing:** Statistical multiplexing is a communication

link sharing technique, which is used in packet switching. The shared linking is variable in statistical multiplexing, whereas it is fixed in TDM or FDM. This is a strategic application for maximizing the utilization of bandwidth. This can increase the efficiency of network, as well. Statistical multiplexing is based on the principle of demand and availability. The signals are transmitted over the medium only when they have data to send and when the medium is available. The signals are separated at the receiver by using headers. Statistical multiplexing is suitable for bursty and unpredictable traffic and can be used in internet and wireless networks.

A simple mnemonic to remember the types of switching techniques is:

- Circuit switching: **C**onstant path, **C**onstant bandwidth, **C**ostly
- Message switching: **M**essage as a unit, **M**emory required, **M**ore delay
- Packet switching: **P**acket as a unit, **P**ath varies, **P**erformance better

A simple mnemonic to remember the types of multiplexing is:

- FDM: **F**requency bands, **F**ilters, **F**or analog signals
- TDM: **T**ime slots, **T**iming, **T**or digital signals
- Statistical multiplexing: **S**hared link, **S**tatistics, **S**uitable for packet switching

Unit 2 - Link layer in Computer Networks and Medium Access Control and Local Area Networks

- The link layer is the second layer of the OSI model of computer networking. It is responsible for transferring data between nodes on a network segment across the physical layer.
- The link layer is divided into two sublayers: the logical link control (LLC) sublayer and the medium access control (MAC) sublayer.
- The LLC sublayer provides an abstraction of the physical layer to the upper layers of the OSI model. It also performs error control, flow control, and framing of data packets.
- The MAC sublayer is responsible for managing the access to the shared physical medium among multiple nodes. It also performs addressing, channel allocation, and collision avoidance.
- There are different types of link layer protocols depending on the type of network, such as Ethernet, Wi-Fi, Bluetooth, etc. Each protocol has its own MAC sublayer protocol that defines how the nodes communicate with each other.
- A local area network (LAN) is a network that connects devices within a limited geographic area, such as a home, office, or campus. A LAN can be organized in four different ways from a physical layer perspective: a bus-shaped network, a ring-shaped network, a star-shaped network, or a mesh-shaped network.
- A bus-shaped network is a network where all nodes are attached to the same physical cable. A ring-shaped network is a network where the nodes

are connected in a circular fashion. A star-shaped network is a network where the nodes are connected to a central hub or switch. A mesh-shaped network is a network where the nodes are connected to each other in a flexible way.

- The MAC sublayer protocols for LANs can be classified into two categories: contention-based protocols and reservation-based protocols.
- Contention-based protocols are protocols where the nodes compete for the access to the channel. They use a random access method, such as ALOHA, CSMA, or CSMA/CD, to transmit data. These protocols are simple, but they may suffer from collisions and low efficiency.
- Reservation-based protocols are protocols where the nodes reserve the access to the channel in advance. They use a deterministic access method, such as TDMA, FDMA, or CDMA, to transmit data. These protocols are more complex, but they can avoid collisions and achieve higher efficiency.
- A mnemonic to remember the difference between contention-based and reservation-based protocols is: Contention-based protocols use **C**hance, while **R**eservation-based protocols use **R**ules.

Link layer in Computer Networks

- The link layer is the lowest layer in the Internet protocol suite, the networking architecture of the Internet.
- The link layer is the group of methods and communications protocols confined to the link that a host is physically connected to.
- The link layer is responsible for transferring data between nodes on a network segment across the physical layer.
- The link layer performs the following functions:
 - Framing: The packet received from the network layer is known as a frame in the link layer. The link layer encapsulates the frame with a header and a trailer that contain control information such as source and destination addresses, error detection codes, and frame sequence numbers.
 - Addressing: The link layer encapsulates the source and destination's MAC address / physical address in the frame header. The MAC address is a unique identifier for each node on the network and is used to deliver the frame to the correct destination.
 - Error Control: The link layer detects and possibly corrects errors that can occur in the physical layer. The link layer uses error detection codes such as parity bits, checksums, or cyclic redundancy checks (CRC) to check the integrity of the frame. The link layer may also use error correction codes such as Hamming codes or Reed-Solomon codes to correct the errors without retransmission. The link layer may also use retransmission schemes such as stop-and-wait, go-back-N, or selective repeat to recover from lost or corrupted frames.
 - Flow Control: The link layer regulates the flow of data between the sender and the receiver to avoid congestion or overflow. The link layer

may use feedback-based schemes such as sliding window or rate-based schemes such as leaky bucket or token bucket to control the flow of data.

- Media Access Control: The link layer coordinates the access of multiple nodes to the shared medium or channel. The link layer may use multiple access protocols such as ALOHA, CSMA, CSMA/CA, CSMA/CD, or TDMA to avoid or resolve collisions and ensure fair and efficient use of the channel.
- The link layer may be implemented in hardware, software, or a combination of both. The link layer may also be divided into sublayers such as logical link control (LLC) and media access control (MAC) to provide different services and interfaces to the upper and lower layers.
- The link layer is also known as the data link layer or layer 2 in the OSI model of computer networking.
- A mnemonic to remember the functions of the link layer is **FAME** (Framing, Addressing, Media Access Control, Error Control). A mnemonic to remember the multiple access protocols is **ACT** (ALOHA, CSMA, TDMA).

Framing in link layer in Computer Networks

- Framing is a function of the data link layer. It provides a way for a sender to transmit a set of bits that are meaningful to the receiver .
- Framing uses frames to send or receive data. The data link layer receives packets from the network layer and converts them into frames .
- Frames have headers that contain information such as error-checking codes, source and destination addresses, and control information .
- Frames are the result of the final layer of encapsulation before the data is transmitted over the physical layer.
- Framing is necessary because the physical layer only accepts and transfers a stream of bits without any regard to meaning or structure .
- Framing also helps to synchronize the transmission and reception of data, and to detect and correct errors .
- There are different types of framing methods, such as:
 - Character count: The header contains a field that specifies the number of characters in the frame. This method is simple but unreliable, as any error in the count field or the frame data can cause loss of synchronization .
 - Byte stuffing: The frame starts and ends with a special byte pattern, such as DLE STX (Data Link Escape, Start of Text) and DLE ETX (End of Text). If the frame data contains the same pattern, it is stuffed with an extra DLE byte to avoid confusion. This method is

more reliable but requires extra overhead for stuffing and destuffing .

- Bit stuffing: The frame starts and ends with a special bit pattern, such as 01111110. If the frame data contains five consecutive 1s, a 0 is stuffed after them to avoid confusion. This method is also reliable but requires extra overhead for stuffing and destuffing .
 - Physical layer coding violations: The frame uses the physical layer encoding scheme, such as Manchester or NRZ, to mark the start and end of the frame. For example, a violation of the Manchester scheme, such as 00 or 11, can indicate the frame boundary. This method is efficient but depends on the physical layer implementation .
- A possible mnemonic to remember the four framing methods is: **Charlie Brown Bites Peanuts**, where C stands for Character count, B stands for Byte stuffing and Bit stuffing, and P stands for Physical layer coding violations.
 - A possible ascii diagram to illustrate the framing process is:

Network layer packet: | DATA |

Data link layer frame: | HEADER | DATA | TRAILER |

Physical layer bits: 0101010101010101 | HEADER | DATA | TRAILER | 0101010101010101

Error Detection and Correction in link layer in Computer Networks

- Error detection and correction in link layer is the process of detecting and correcting data frames that have been corrupted or lost during transmission .
- The link layer uses various error control techniques for this purpose. The basic approach is the use of redundancy, where additional bits are added to facilitate the detection and correction of errors .
- There are three main techniques for detecting errors in frames: Parity Check, Checksum and Cyclic Redundancy Check (CRC) .
 - Parity Check: The parity check is done by adding an extra bit, called parity bit to the data to make a number of 1s either even in case of even parity or odd in case of odd parity. For example, if the data is 1011 and even parity is used, the parity bit will be 0 to make the total number of 1s even. If odd parity is used, the parity bit will be 1 to make the total number of 1s odd. The receiver can check the parity bit and detect if there is a single bit error in the data. However, this technique cannot detect multiple bit errors or burst errors.
 - Checksum: The checksum is a value calculated from the data by applying some mathematical function, such as addition or XOR. The sender appends the checksum to the data and sends it to the receiver.

The receiver recalculates the checksum from the received data and compares it with the received checksum. If they match, the data is assumed to be error-free. Otherwise, there is an error in the data. For example, if the data is 1011 and 1100, the checksum can be the sum of the two values, which is 10111. The receiver can add the received data and compare it with the received checksum. However, this technique cannot detect errors that cancel out each other, such as 1011 and 0100.

- Cyclic Redundancy Check (CRC): The CRC is a value calculated from the data by applying a polynomial function, such as $x^3 + x + 1$. The sender appends the CRC to the data and sends it to the receiver. The receiver divides the received data by the same polynomial and checks the remainder. If the remainder is zero, the data is assumed to be error-free. Otherwise, there is an error in the data. For example, if the data is 1011 and the polynomial is $x^3 + x + 1$, the CRC can be the remainder of dividing 1011000 by the polynomial, which is 011. The receiver can divide the received data by the same polynomial and compare the remainder with the received CRC. However, this technique cannot detect errors that are multiples of the polynomial.
- Error correction in link layer is the process of recovering the original data from the corrupted or lost data frames. There are two main techniques for error correction in link layer: Automatic Repeat Request (ARQ) and Forward Error Correction (FEC) .
 - Automatic Repeat Request (ARQ): The ARQ is a technique that uses feedback from the receiver to the sender to request retransmission of corrupted or lost data frames. The sender keeps a copy of the sent data frames until it receives an acknowledgment (ACK) from the receiver. The receiver sends an ACK for each error-free data frame it receives. If the sender does not receive an ACK within a certain time, it assumes that the data frame was corrupted or lost and retransmits it. There are three main types of ARQ: Stop-and-Wait ARQ, Go-Back-N ARQ and Selective Repeat ARQ.
 - * Stop-and-Wait ARQ: The Stop-and-Wait ARQ is a technique that allows the sender to send only one data frame at a time and wait for an ACK before sending the next one. The receiver sends an ACK for each data frame it receives. If the sender does not receive an ACK within a certain time, it retransmits the same data frame. This technique is simple but inefficient, as it wastes a lot of bandwidth and time by waiting for ACKs.
 - * Go-Back-N ARQ: The Go-Back-N ARQ is a technique that allows the sender to send multiple data frames without waiting for ACKs, up to a certain window size. The receiver sends an ACK for

Flow control in link layer in Computer Networks

- Flow control is a technique that allows two stations working at different speeds to communicate with each other.
- It regulates the amount of data that a sender can send before receiving an acknowledgment from the receiver .
- It prevents the sender from overwhelming the receiver with too many frames or data.
- There are two main methods of flow control in the link layer: stop-and-wait and sliding window.

Stop-and-wait

- In this method, the sender sends one frame at a time and waits for an acknowledgment from the receiver before sending the next frame.
- The sender can only send one frame in each transmission round.
- The receiver sends an acknowledgment after receiving and processing each frame.
- The acknowledgment can be positive (ACK) or negative (NAK) depending on whether the frame was received correctly or not.
- If the sender receives a positive acknowledgment, it sends the next frame. If it receives a negative acknowledgment or no acknowledgment within a timeout period, it retransmits the same frame.
- The advantage of this method is that it is simple and reliable.
- The disadvantage of this method is that it is inefficient and slow, especially when the transmission delay is large compared to the processing time.

Sliding window

- In this method, the sender can send multiple frames in each transmission round without waiting for acknowledgments.
- The sender maintains a window of frames that it can send at any time. The size of the window is determined by the receiver's buffer capacity.
- The receiver also maintains a window of frames that it can receive at any time. The size of the window is determined by the sender's window size and the sequence numbers of the frames.
- The receiver sends an acknowledgment after receiving and processing each frame. The acknowledgment indicates the next expected frame and the current window size.
- The sender slides its window forward by the number of frames acknowledged by the receiver. The sender can send new frames that fall within the window.
- The receiver slides its window forward by the number of frames received correctly. The receiver can accept new frames that fall within the window.
- The advantage of this method is that it is efficient and fast, especially when the transmission delay is small compared to the processing time.
- The disadvantage of this method is that it is complex and requires more buffer space and sequence numbers.

Mnemonics and learning tricks

- To remember the difference between stop-and-wait and sliding window, you can use the following mnemonics:
 - Stop-and-wait: One frame at a time, wait for ACK or NAK, simple and slow.
 - Sliding window: Multiple frames at a time, slide the window, complex and fast.
- To remember the formula for the efficiency of stop-and-wait, you can use the following trick:
 - Efficiency = $1 / (1 + 2a)$, where a = transmission delay / processing time.
 - The numerator is 1 because the sender can only send one frame in each round.
 - The denominator is $1 + 2a$ because the sender has to wait for the round-trip time of the frame and the acknowledgment, which is twice the transmission delay, plus the processing time of the receiver.
 - The efficiency is inversely proportional to the transmission delay and directly proportional to the processing time.

Elementary Data Link Protocols in link layer in Computer Networks

- Protocols in the data link layer are designed to perform the basic functions of framing, error control and flow control.
- Framing is the process of dividing bit-streams from physical layer into data frames whose size ranges from a few hundred to a few thousand bytes.
- Error control is the process of detecting and correcting errors that may occur during transmission or reception of data frames.
- Flow control is the process of regulating the rate of data transmission between the sender and the receiver to avoid congestion or buffer overflow.
- Elementary data link protocols are classified into three categories, as given below:
 - Protocol 1: Unrestricted simplex protocol
 - Protocol 2: Simplex stop-and-wait protocol
 - Protocol 3: Simplex protocol for noisy channels.
- Let us discuss each protocol one by one.

Protocol 1: Unrestricted simplex protocol

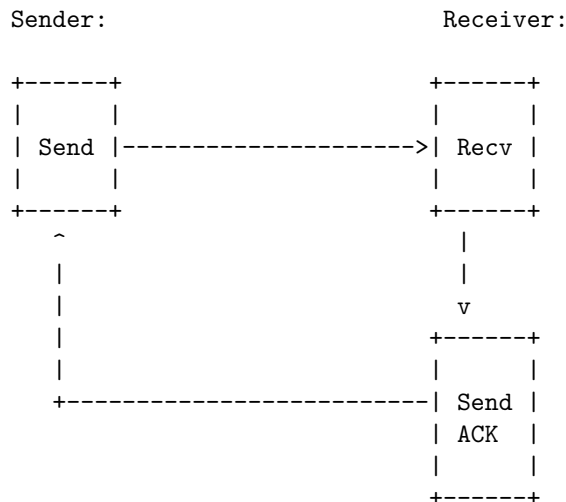
- This protocol is used for noiseless channels, that is, channels that do not introduce any errors or losses in the data frames.
- In this protocol, the sender can send data frames continuously without waiting for any acknowledgment or feedback from the receiver.

- ```

Sender: Receiver:
+-----+ +-----+
Send	----->	Recv
+-----+ +-----+

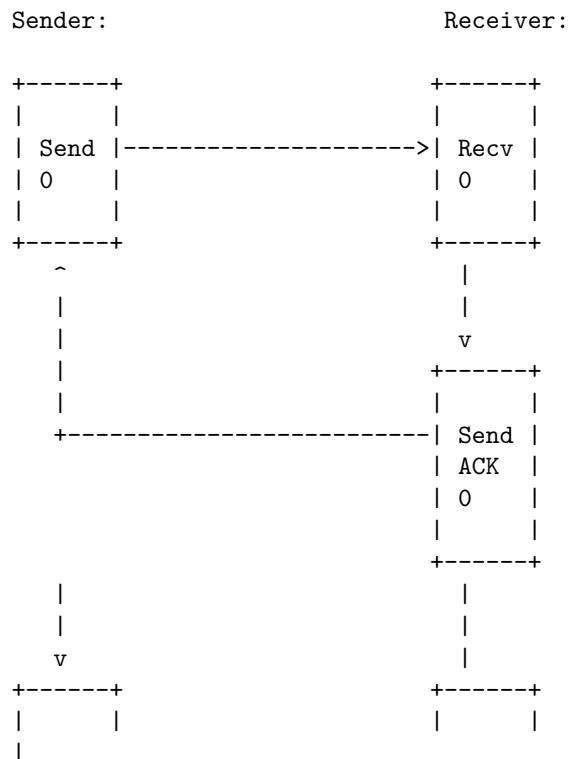
```

- This protocol is used for noisy channels, that is, channels that may introduce errors or losses in the data frames.
- In this protocol, the sender sends one data frame and waits for an acknowledgment (ACK) from the receiver before sending the next data frame.
- The receiver sends an ACK after receiving and processing a data frame correctly.
- If the sender does not receive an ACK within a specified time-out period, it assumes that the data frame or the ACK was lost or corrupted, and retransmits the data frame.
- This protocol has error control but no flow control mechanisms.
- This protocol is reliable but inefficient, as the sender has to wait for each ACK, which reduces the channel utilization and throughput.
- This protocol can be represented by the following state diagram:



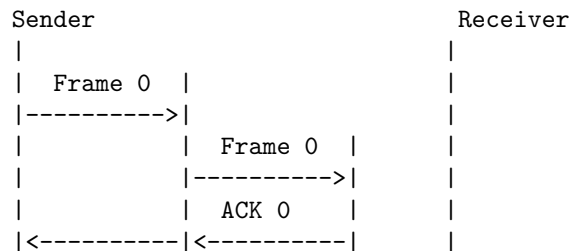
### Protocol 3: Simplex protocol for noisy channels

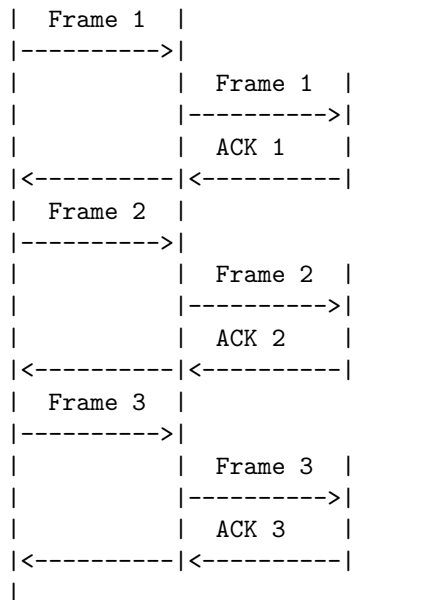
- This protocol is an improvement over protocol 2, as it introduces a sequence number for each data frame and ACK to avoid the problem of duplicate frames.
- In this protocol, the sender and the receiver agree on a window size of one, that is, the sender can have only one outstanding (unacknowledged) data frame at a time.
- The sender attaches a sequence number (0 or 1) to each data frame and alternates between them for successive frames.
- The receiver checks the sequence number of the incoming data frame and sends an ACK with the same sequence number if the frame is correct and in order.
- The sender expects an ACK with the same sequence number as the data frame it sent. If it receives an ACK with a different sequence number, it discards it and retransmits the data frame.
- This protocol has error control and flow control mechanisms, but the window size of one limits the channel utilization and throughput.
- This protocol can be represented by the following state diagram:



## Sliding Window protocols in link layer in Computer Networks

- The sliding window protocol is a data link layer protocol that is useful in the sequential and reliable delivery of the data frames .
- Using the sliding window protocol, the sender can send multiple frames at a time before receiving an acknowledgment from the receiver .
- The sliding window is also used in Transmission Control Protocol (TCP), which operates at the transport layer .
- The sliding window protocol manages the flow of data between two network nodes to ensure that the receiver can handle the incoming data and that the sender does not overwhelm the network .
- The sliding window protocol uses two types of windows: a sender window and a receiver window .
- The sender window is the set of frames that the sender can send without waiting for an acknowledgment .
- The receiver window is the set of frames that the receiver can accept without sending an acknowledgment .
- The size of the windows can vary depending on the protocol and the network conditions .
- The sender and the receiver use sequence numbers to identify and acknowledge the frames .
- The sender and the receiver also use timers to detect and retransmit lost or corrupted frames .
- There are two main variants of the sliding window protocol: stop-and-wait and go-back-N .
- In stop-and-wait, the sender sends one frame at a time and waits for an acknowledgment before sending the next frame .
- In go-back-N, the sender can send up to N frames at a time without waiting for an acknowledgment, where N is the size of the sender window .
- The stop-and-wait protocol is simple but inefficient, as it wastes the network bandwidth and causes long delays .
- The go-back-N protocol is more efficient but more complex, as it requires the sender to keep track of the frames in the window and retransmit them if they are lost or corrupted .
- A mnemonic to remember the difference between stop-and-wait and go-back-N is: stop-and-wait is like a **single-lane** road, where only one car can pass at a time, and go-back-N is like a **multi-lane** road, where multiple cars can pass at a time.
- A diagram to illustrate the sliding window protocol is:





- Some advantages of the sliding window protocol are :
  - It improves the network efficiency and throughput by allowing multiple frames to be sent at a time.
  - It ensures the reliable and sequential delivery of the data frames by using sequence numbers and acknowledgments.
  - It prevents the network congestion and buffer overflow by controlling the flow of data between the

## Medium Access Control and Local Area Networks

- Medium Access Control (MAC) is a sublayer of the data link layer that defines how multiple devices share the same medium (such as a cable or a wireless channel) in a local area network (LAN).
- MAC protocols are responsible for controlling the access to the medium, avoiding or resolving collisions, and ensuring reliable and efficient transmission of data.
- There are two main approaches to MAC in LANs: contention and token-passing.
  - Contention is a first-come, first-serve approach, where each device tries to access the medium whenever it has data to send. If two or more devices access the medium at the same time, a collision occurs and the devices have to retransmit their data after a random delay.
  - Token-passing is a controlled access approach, where a special message called a token is passed among the devices in a predefined order. Only the device that holds the token can access the medium and send data. After sending data, the device passes the token to the

next device in the order.

- The most used contention-based MAC protocol is Carrier Sense Multiple Access/Collision Detection (CSMA/CD), used in Ethernet networks. CSMA/CD works as follows:
  - Before sending data, a device senses the medium to check if it is idle or busy. If the medium is idle, the device starts transmitting data. If the medium is busy, the device waits until it becomes idle.
  - While sending data, a device monitors the medium for any signs of collision. If a collision is detected, the device stops transmitting and sends a jamming signal to inform other devices of the collision. Then, the device waits for a random time and tries to retransmit the data.
- The most used token-passing MAC protocol is Token Ring, used in some IBM networks. Token Ring works as follows:
  - The devices are connected in a logical ring topology, where each device has a successor and a predecessor. A token is circulated among the devices in one direction.
  - When a device receives the token, it checks if it has data to send. If it has data, it sets a busy bit in the token and sends the data along with the token. If it has no data, it passes the token to the next device without changing the busy bit.
  - When a device receives a data frame along with the token, it checks if the frame is addressed to it. If it is, it copies the frame and sets an acknowledgment bit in the token. If it is not, it passes the frame and the token to the next device without changing the acknowledgment bit.
  - When the sender receives the frame and the token back, it checks the acknowledgment bit. If it is set, it means the frame was received successfully and the sender can release the token by clearing the busy bit. If it is not set, it means the frame was lost or corrupted and the sender can retransmit the frame by keeping the busy bit.
- A mnemonic to remember the difference between contention and token-passing is: **C**ontention is **C**haotic, **T**oken-passing is **T**idy.

### Channel allocation in medium access control

- Channel allocation is the process of assigning channels (frequency bands, time slots, code sequences, etc.) to different users or nodes in a communication network.
- Medium access control (MAC) is the protocol or mechanism that coordinates the access to a shared channel by multiple users or nodes, while ensuring good performance, fairness, low latency and low energy consumption.
- Channel allocation and medium access control are closely related, as the MAC protocol may depend on the channel allocation scheme, and vice versa.



- There are three main types of channel allocation schemes: fixed, dynamic and hybrid.
  - Fixed channel allocation (FCA) assigns a fixed number of channels to each user or node, regardless of the traffic demand or channel conditions. FCA is simple and efficient, but may suffer from under-utilization or congestion of channels.
  - Dynamic channel allocation (DCA) assigns channels to users or nodes on demand, based on the traffic load, channel quality, priority, etc. DCA is flexible and adaptive, but may incur higher overhead and complexity.
  - Hybrid channel allocation (HCA) combines the advantages of FCA and DCA, by dividing the channels into two groups: one for fixed allocation and one for dynamic allocation. HCA can balance the trade-off between efficiency and flexibility, but may require more coordination and synchronization.
- There are two main types of MAC protocols: contention-based and contention-free.
  - Contention-based MAC protocols allow users or nodes to compete for the channel access, using random or deterministic methods. Contention-based MAC protocols are simple and decentralized, but may suffer from collisions, delays and unfairness.
  - Contention-free MAC protocols allocate the channel access to users or nodes in a predetermined or negotiated manner, using polling, reservation, token passing, etc. Contention-free MAC protocols are collision-free and predictable, but may incur higher overhead and complexity.
- A mnemonic to remember the types of channel allocation schemes and MAC protocols is:
  - FCA: Fixed Channel Allocation
  - DCA: Dynamic Channel Allocation
  - HCA: Hybrid Channel Allocation
  - CBM: Contention-Based MAC
  - CFM: Contention-Free MAC
- A learning trick to understand the difference between channel allocation and medium access control is:
  - Channel allocation is like assigning seats to passengers in a plane, while medium access control is like managing the boarding process.

### **Multiple access protocols in medium access control**

- Multiple access protocols are a set of protocols operating in the Medium Access Control sublayer (MAC sublayer) of the Open Systems Intercon-

nection (OSI) model .

- These protocols allow a number of nodes or users to access a shared network channel, such as a wireless channel or a bus network, without interfering with each other or causing collisions .
- Multiple access protocols can be classified into three main categories: random access, controlled access, and channelization.
  - Random access protocols: In these protocols, all stations have equal priority and can send data depending on the medium's state (idle or busy). There is no fixed time for sending data and no central controller to coordinate the transmissions. Examples of random access protocols are ALOHA, Carrier Sense Multiple Access (CSMA), CSMA with Collision Avoidance (CSMA/CA), and CSMA with Collision Detection (CSMA/CD) .
  - Controlled access protocols: In these protocols, there is a central controller or a predefined rule that determines which station can access the channel at a given time. The stations have to request permission or wait for their turn to send data. Examples of controlled access protocols are Reservation, Polling, and Token Passing.
  - Channelization protocols: In these protocols, the channel is divided into smaller subchannels or time slots that are assigned to different stations. The stations can only send data in their assigned subchannels or time slots, and there is no contention or collision among them. Examples of channelization protocols are Frequency Division Multiple Access (FDMA), Time Division Multiple Access (TDMA), and Code Division Multiple Access (CDMA).
- The choice of a multiple access protocol depends on various factors, such as the type of network, the traffic characteristics, the channel conditions, the performance requirements, and the cost and complexity of implementation.
- Some advantages of multiple access protocols are:
  - They enable efficient and fair use of the shared channel by multiple stations .
  - They reduce the delay and overhead of data transmission by avoiding or minimizing collisions and retransmissions .
  - They adapt to the dynamic and unpredictable nature of network traffic and channel conditions by using feedback or sensing mechanisms .
- Some disadvantages of multiple access protocols are:
  - They introduce additional complexity and overhead in the MAC sublayer, such as framing, error control, synchronization, and coordination .
  - They may not guarantee the quality of service (QoS) or the fairness of access for all stations, especially in high traffic or noisy scenarios .
  - They may suffer from performance degradation or instability due to channel errors, hidden terminals, exposed terminals, or capture effects .

- A possible mnemonic to remember the three categories of multiple access protocols is: **RCC** (Random, Controlled, Channelization). A possible learning trick to remember the examples of each category is: **APCT** (ALOHA, Polling, CDMA, Token Passing). The first letter of each example corresponds to the first letter of the category it belongs to.

### LAN standards in local area network

- A local area network (LAN) is a data communication network connecting various terminals or computers within a building or limited geographical area. The connection among the devices could be wired or wireless.
- LAN standards are the specifications and protocols that define the physical and logical aspects of how devices communicate and share resources on a LAN. LAN standards are developed and maintained by various organizations, such as the Institute of Electrical and Electronics Engineers (IEEE), the American National Standards Institute (ANSI), and the International Organization for Standardization (ISO).
- The most widely used and recognized LAN standards are the IEEE 802 family of standards, which cover different types of LAN technologies, such as Ethernet, Token Ring, Wireless LAN, and others . Each IEEE 802 standard has a number and a letter that indicates the type of LAN technology and the focus group that developed it. For example, IEEE 802.3 is the standard for Ethernet, and IEEE 802.11 is the standard for Wireless LAN.
- Some of the main features and characteristics of the IEEE 802 LAN standards are:
  - They use a common logical link control (LLC) sublayer that provides a uniform interface between the upper layers of the network stack and the different media access control (MAC) sublayers of the various LAN technologies.
  - They support different types of network topologies, such as bus, star, ring, and mesh, depending on the LAN technology and the physical medium used .
  - They define different methods of media access control, such as contention-based (e.g., CSMA/CD for Ethernet), token-based (e.g., Token Ring), and polling-based (e.g., FDDI), to coordinate the transmission of frames among the devices on the LAN .
  - They specify different frame formats, addressing schemes, error detection and correction mechanisms, and other parameters for each LAN technology .
  - They support different data rates, ranging from 1 Mbps to 10 Gbps or more, depending on the LAN technology and the physical medium used .

- They enable interoperability and compatibility among devices from different vendors that comply with the same LAN standard .
- A mnemonic to remember the main IEEE 802 LAN standards and their corresponding LAN technologies is:
  - **E**very **T**ime **W**e **T**alk, **W**e **P**lay **M**usic **R**eally **L**oudly
  - **E**thernet (802.3), **T**oken Ring (802.5), **W**ireless LAN (802.11), **T**oken Bus (802.4), **W**ireless PAN (802.15), **P**owerline Communication (802.21), **M**etropolitan Area Network (802.16), **R**esilient Packet Ring (802.17), **L**ocal and Metropolitan Area Network (802.1)

### Link layer switches & bridges in local area network

- Link layer switches and bridges are network devices that operate at the data link layer (layer 2) of the OSI model and connect multiple LANs (local area networks) together to form a larger LAN.
- They use MAC addresses to forward Ethernet frames from one device to another device in an Ethernet standard based LAN.
- They store (store-and-forward mode) and then regenerate the frame, which helps to reduce errors and collisions in the network.
- They can also interconnect data link layer domains that have different technologies, such as Ethernet to FDDI, by converting the frame format and adjusting the maximum frame size.
- Link layer switches are more advanced than bridges, as they can have multiple ports and perform filtering, learning, and forwarding functions based on MAC addresses.
- Link layer switches can also support VLANs (virtual LANs), which are logical groups of devices that share the same broadcast domain, regardless of their physical location or connection.
- Link layer switches and bridges can improve the performance, scalability, and security of a LAN.

Some mnemonics and learning tricks for link layer switches and bridges are:

- Switches and bridges work at layer 2, which rhymes with “glue”. They glue multiple LANs together to form a bigger LAN.
- Switches and bridges use MAC addresses, which are also called physical addresses. They deal with the physical devices in the network, not the logical ones.
- Switches and bridges store and forward frames, which sounds like “store and reward”. They reward the network by reducing errors and collisions.
- Switches and bridges can interconnect different technologies, such as Ethernet and FDDI. They are like translators that can speak different languages and convert them to a common one.

### Learning bridge algorithms in local area network

- A bridge is a device that connects two or more local area networks (LANs) at the data link layer and forwards frames based on the destination MAC address.
- A bridge algorithm is a method that determines how a bridge learns and updates its forwarding table and how it handles loops and broadcast storms in the network.
- There are two main types of bridge algorithms: spanning tree (ST) and source routing (SR), which are specified by the IEEE 802 standards committee .
- Spanning tree algorithm:
  - It creates a loop-free logical topology of the network by disabling some of the bridge ports and forming a tree of active ports.
  - It uses a distributed protocol that elects a root bridge and assigns a cost to each port based on the bandwidth and distance to the root bridge.
  - It updates the forwarding table by learning the MAC addresses from the incoming frames and aging out the entries that are not used for a certain period of time.
  - It handles topology changes by sending and receiving special frames called bridge protocol data units (BPDUs) that inform other bridges about the new status of the ports.
  - It has the advantages of simplicity, transparency, and compatibility with any LAN technology.
  - It has the disadvantages of slow convergence, suboptimal routing, and wasted bandwidth on disabled ports.
- Source routing algorithm:
  - It embeds the route information in the frame header and lets the source node decide the path to the destination node.
  - It uses a discovery mechanism that sends special frames called explorer frames to find all possible routes to the destination node and stores them in a routing information field (RIF).
  - It updates the forwarding table by caching the RIFs from the incoming frames and discarding the entries that are not used for a certain period of time.
  - It handles topology changes by sending and receiving special frames called ring purge frames (RPFs) that inform other nodes about the new status of the links.
  - It has the advantages of fast convergence, optimal routing, and load balancing.
  - It has the disadvantages of complexity, overhead, and incompatibility with some LAN technologies.
- A mnemonic to remember the difference between ST and SR is: ST is Simple and Transparent, SR is Smart and Resourceful.

## Spanning Tree Algorithms in Local Area Network

- Spanning tree algorithms are used to prevent loops in a network topology that uses bridges or switches to connect multiple segments of a local area network (LAN).
- Loops can cause problems such as broadcast storms, multiple frame copies, and MAC address table instability.
- Spanning tree algorithms work by creating a logical tree structure of the network, where only one path exists between any two nodes. This path is called the spanning tree.
- The spanning tree is determined by electing a root bridge, which is the switch with the lowest bridge ID. The bridge ID consists of a priority value and a MAC address. The default priority value is 32768 and can be changed by the network administrator.
- The root bridge sends out special frames called bridge protocol data units (BPDUs) to announce its presence and status to other switches. The BPDUs contain information such as the root bridge ID, the sender bridge ID, the cost to reach the root bridge, and the port ID.
- The other switches receive the BPDUs and compare them to their own information. They update their root bridge, root port, and designated port information accordingly. The root port is the port that has the lowest cost to reach the root bridge. The designated port is the port that has the lowest cost to reach a segment. The cost is based on the bandwidth of the link.
- The switches forward the BPDUs to their other ports, except for the root port and the blocked ports. The blocked ports are the ports that are not part of the spanning tree and are disabled to prevent loops.
- The spanning tree algorithm runs continuously to detect any changes in the network topology, such as link failures, new switches, or configuration changes. If a change occurs, the switches exchange BPDUs and recalculate the spanning tree.
- There are different versions of spanning tree protocols, such as the original IEEE 802.1D standard, the Rapid Spanning Tree Protocol (RSTP), the Multiple Spanning Tree Protocol (MSTP), and the Cisco proprietary Per-VLAN Spanning Tree Protocol (PVST+). They differ in terms of speed, efficiency, and scalability.

### Unit 3 - Network Layer in Computer Networks

- The network layer is the third layer in the OSI model of computer networks. It is also called the Internet layer in the TCP/IP model.
- The main function of the network layer is to transfer network packets from the source to the destination across different networks or subnets.
- The network layer provides the following services to the transport layer and the upper layers:
  - Host and network addressing: The network layer assigns logical addresses (such as IP addresses) to the hosts and networks, and uses them to identify and locate the source and destination of the packets.

- Routing: The network layer determines the best path or route for the packets to reach the destination, based on various factors such as distance, cost, congestion, etc. The network layer uses routers to forward the packets across different networks or subnets.
- Fragmentation and reassembly: The network layer may divide a large packet into smaller fragments to fit the maximum transmission unit (MTU) of the underlying network. The network layer also reassembles the fragments at the destination and delivers the original packet to the transport layer.
- Error control and congestion control: The network layer may detect and correct errors in the packets, such as checksum errors, and retransmit the corrupted packets. The network layer may also control the flow of packets to avoid congestion or overload in the network.
- Quality of service (QoS): The network layer may provide different levels of service to different types of packets, such as delay, bandwidth, reliability, etc. The network layer may use different techniques such as priority, reservation, or differentiation to achieve QoS.
- The network layer can be divided into two sublayers: the network layer protocol and the network layer interface.
  - The network layer protocol is the set of rules and formats that define how the packets are structured, addressed, routed, and delivered. Examples of network layer protocols are IP, ICMP, ARP, RARP, etc.
  - The network layer interface is the set of functions and procedures that allow the network layer to communicate with the data link layer and the transport layer. The network layer interface may use different types of sockets, such as TCP sockets, UDP sockets, or raw sockets, to send and receive packets.
- The network layer can be implemented in different ways, such as connectionless or connection-oriented, datagram or virtual circuit, stateless or stateful, etc. These design choices affect the performance, reliability, and complexity of the network layer.
  - Connectionless network layer: The network layer does not establish or maintain any connection or session between the source and destination. Each packet is treated independently and may take different routes or arrive out of order. The network layer does not guarantee the delivery, order, or integrity of the packets. The transport layer or the application layer is responsible for error control and reordering. Examples of connectionless network layer protocols are IP and ICMP.
  - Connection-oriented network layer: The network layer establishes and maintains a connection or session between the source and destination before sending any packets. The network layer reserves resources and assigns a unique identifier to the connection. The network layer guarantees the delivery, order, and integrity of the packets. The network layer is responsible for error control and reordering. Examples of connection-oriented network layer protocols are X.25 and ATM.

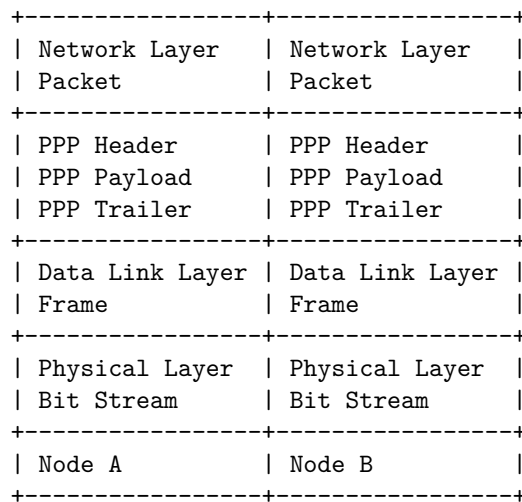
- Datagram network layer: The network layer treats each packet as a self-contained unit of information, with its own source and destination addresses. The network layer does not keep any information about the packets or the connections. The network layer is simple, flexible, and scalable, but may suffer from congestion, delay, or loss. Examples of datagram network layer protocols are IP and ICMP.
- Virtual circuit network layer: The network layer establishes a logical connection or circuit between the source and destination, and assigns a label or number to each packet. The network layer keeps information about the packets and the connections in a table or database. The network layer is more reliable, efficient, and secure, but may be complex, rigid, and costly. Examples of virtual circuit network layer protocols are X.25 and ATM.
- Stateless network layer: The network layer does not store any information about the packets or the connections. The network layer does not keep track of the state or status of the packets or the connections. The network layer is simple, fast, and scalable, but may not be able to handle errors, congestion, or QoS. Examples of stateless network layer protocols are IP and ICMP.
- Stateful network layer: The network layer stores information about the packets or the connections. The network layer keeps track of the state or

### **Point-to-point networks in network layer**

- A point-to-point network is a network topology that connects two nodes directly using a single link.
- A point-to-point link can be either physical (such as a cable) or logical (such as a tunnel).
- A point-to-point network can be used to connect two routers, two hosts, or a router and a host.
- A point-to-point network can provide dedicated bandwidth, security, and reliability for the communication between the two nodes.
- A point-to-point network can use different protocols at the data link layer (layer 2) to encapsulate the network layer (layer 3) packets for transmission over the link.
- One of the most common protocols for point-to-point networks is the Point-to-Point Protocol (PPP), which can provide authentication, encryption, and compression features.
- PPP consists of three main components:
  - A way to encapsulate multiprotocol datagrams into frames.
  - A Link Control Protocol (LCP) to establish, configure, and test the data link connection.
  - A set of Network Control Protocols (NCPs) to negotiate the network layer parameters for different protocols (such as IP, IPX, or AppleTalk).



- PPP can operate over various types of physical media, such as serial ports, modems, ISDN, Ethernet, or wireless.
- PPP can also operate over logical media, such as Point-to-Point Tunneling Protocol (PPTP), which creates a virtual point-to-point link over an IP network.
- A mnemonic to remember the components of PPP is **ELN** (Encapsulation, Link Control, Network Control).
- A diagram to illustrate the structure of PPP is:



### Logical addressing in network layer

- Logical addressing is a way of identifying devices on a network using a logical name or number, rather than a physical address.
- Logical addresses are assigned by the network layer protocol, such as IP (Internet Protocol), and are independent of the underlying data link layer or physical layer addresses, such as MAC (Media Access Control) addresses or serial numbers.
- Logical addresses are used to route packets from the source to the destination across multiple networks or subnets, which may have different data link layer or physical layer technologies.
- Logical addresses are usually hierarchical, meaning that they consist of two parts: a network identifier and a host identifier. The network identifier specifies which network or subnet the device belongs to, and the host identifier specifies which device within that network or subnet.
- Logical addresses are usually represented in a human-readable format, such as dotted decimal notation for IPv4 addresses (e.g., 192.168.1.1) or hexadecimal notation for IPv6 addresses (e.g., 2001:db8::1).
- Logical addresses are mapped to physical addresses by the network layer protocol using a mechanism called address resolution. For example, IP

uses ARP (Address Resolution Protocol) or NDP (Neighbor Discovery Protocol) to find the MAC address of a device given its IP address.

- Logical addresses can be static or dynamic. Static addresses are manually configured by the network administrator or the device user, and do not change unless modified. Dynamic addresses are automatically assigned by a server or a protocol, such as DHCP (Dynamic Host Configuration Protocol) or SLAAC (Stateless Address Autoconfiguration), and may change over time.
- Logical addresses can be unicast, multicast, or anycast. Unicast addresses identify a single device on the network, and are used for one-to-one communication. Multicast addresses identify a group of devices on the network, and are used for one-to-many communication. Anycast addresses identify multiple devices on the network that provide the same service, and are used for one-to-nearest communication.

### **Basic internetworking in network layer**

- Internetworking is the process of connecting different networks using routers and other devices to form a larger network that can exchange data across different protocols and technologies.
- Internetworking is implemented in the network layer (layer 3) of the OSI model, which is responsible for routing, addressing, and fragmentation of packets across multiple networks.
- The most common example of internetworking is the Internet, which is a global network of networks that uses the Internet Protocol (IP) to communicate.
- There are three main types of internetworking:
  - Internet: A public network that connects millions of networks and devices worldwide using IP and other protocols.
  - Intranet: A private network that connects networks and devices within an organization using IP and other protocols. An intranet may or may not have access to the Internet.
  - Extranet: A private network that connects networks and devices from different organizations using IP and other protocols. An extranet may or may not have access to the Internet.
- Internetworking involves the following components and concepts:
  - Routers: Devices that forward packets between networks based on their destination addresses. Routers operate at the network layer and use routing tables and algorithms to determine the best path for each packet.
  - Data-link layer addresses: Addresses that uniquely identify each physical network interface of a device. Data-link layer addresses are also known as MAC addresses or hardware addresses. They are usually assigned by the manufacturer and are fixed for each device.
  - Network layer addresses: Addresses that uniquely identify each logical network interface of a device. Network layer addresses are also

known as IP addresses or logical addresses. They are usually assigned by the network administrator or a DHCP server and can be changed for each device.

- Address resolution: The process of mapping network layer addresses to data-link layer addresses. Address resolution is performed by protocols such as ARP (Address Resolution Protocol) or NDP (Neighbor Discovery Protocol).
- Packet fragmentation: The process of breaking a large packet into smaller packets that can fit the maximum transmission unit (MTU) of the underlying network. Packet fragmentation is performed by the sender and the intermediate routers and is reassembled by the receiver.
- Packet reassembly: The process of combining the fragments of a packet into the original packet. Packet reassembly is performed by the receiver and requires the fragments to have identification and sequence numbers.
- Packet delivery: The process of sending and receiving packets across an internetwork. Packet delivery can be either connectionless or connection-oriented. Connectionless packet delivery does not require any prior agreement or acknowledgment between the sender and the receiver, such as IP. Connection-oriented packet delivery requires a prior agreement and acknowledgment between the sender and the receiver, such as TCP.

## IP

- IP stands for Internet Protocol, which is a set of rules that govern how data is transmitted and received over the internet.
- An IP address is a numerical label that identifies a device on a network that uses the IP protocol. For example, 192.0.2.1 is an IP address.
- There are two versions of IP: IPv4 and IPv6. IPv4 uses 32-bit addresses, which can support up to 4.3 billion devices. IPv6 uses 128-bit addresses, which can support up to  $3.4 \times 10^{38}$  devices.
- IP addresses can be classified into two types: public and private. Public IP addresses are assigned by internet service providers (ISPs) and are visible to the internet. Private IP addresses are assigned by network administrators and are only visible within a local network.
- IP addresses can also be classified into two modes: static and dynamic. Static IP addresses are fixed and do not change. Dynamic IP addresses are assigned by a server and can change over time.
- IP addresses are used for network interface identification and location addressing. Network interface identification means that each device on a network can be uniquely identified by its IP address. Location addressing means that each IP address can be mapped to a physical location, such as a city, region, or country.
- IP addresses are also used for routing, which is the process of finding the

best path for data packets to travel from a source to a destination. Routers are devices that use IP addresses to forward packets to the next hop on the path.

- IP addresses are written in dotted decimal notation for IPv4 and hexadecimal notation for IPv6. For example, 192.0.2.1 is an IPv4 address and 2001:db8::1 is an IPv6 address.
- A mnemonic to remember the difference between IPv4 and IPv6 is: IPv4 has four numbers, IPv6 has six letters.
- A learning trick to find your IP address on a Windows device is: On the taskbar, select Wi-Fi network > the Wi-Fi network you're connected to > Properties. Under Properties, look for your IP address listed next to IPv4 address or IPv6 address.

## CIDR

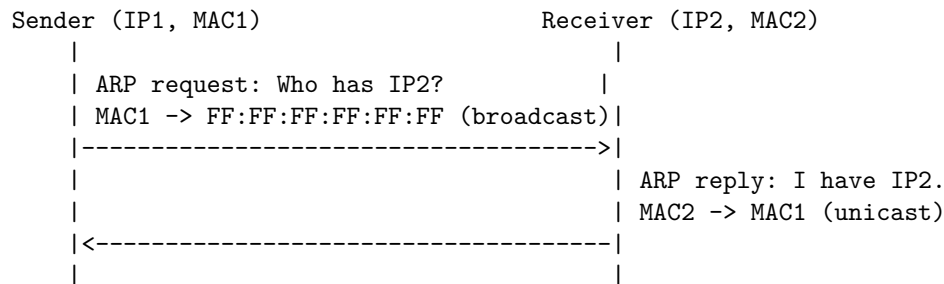
- CIDR stands for Classless Inter-Domain Routing, a method for allocating IP addresses and for IP routing .
- CIDR replaces the previous classful network addressing architecture on the Internet, which was based on fixed classes A, B, and C .
- CIDR allows blocks of IP addresses to be grouped into single routing table entries, which reduces the size and complexity of routing tables and improves the efficiency of address distribution .
- CIDR notation is a compact representation of an IP address and its associated routing prefix. It consists of an IP address, a slash (/), and a number that indicates the number of bits in the prefix .
- For example, 192.168.1.0/24 is a CIDR notation that represents the IP address 192.168.1.0 and its prefix of 24 bits, which corresponds to the subnet mask 255.255.255.0. This means that the network has 256 possible IP addresses, from 192.168.1.0 to 192.168.1.255 .
- CIDR notation can also be used to represent a range of IP addresses, such as 192.168.0.0/16, which covers all the addresses from 192.168.0.0 to 192.168.255.255, or 10.0.0.0/8, which covers all the addresses from 10.0.0.0 to 10.255.255.255 .
- CIDR notation can be converted to IPv4 address range using a utility tool or by applying some simple arithmetic operations. For example, to convert 192.168.1.0/24 to IPv4 address range, we can do the following steps:
  - Find the number of host bits by subtracting the prefix length from 32. In this case,  $32 - 24 = 8$  host bits.
  - Find the number of possible hosts by raising 2 to the power of the host bits. In this case,  $2^8 = 256$  possible hosts.
  - Find the network address by performing a bitwise AND operation between the IP address and the subnet mask. In this case,  $192.168.1.0 \text{ AND } 255.255.255.0 = 192.168.1.0$ .
  - Find the broadcast address by performing a bitwise OR operation between the network address and the inverted subnet mask. In this case,  $192.168.1.0 \text{ OR } 0.0.0.255 = 192.168.1.255$ .

- Find the first and last usable IP addresses by adding 1 to the network address and subtracting 1 from the broadcast address. In this case,  $192.168.1.0 + 1 = 192.168.1.1$  and  $192.168.1.255 - 1 = 192.168.1.254$ .
- The IPv4 address range is then the first and last usable IP addresses, separated by a dash. In this case, 192.168.1.1-192.168.1.254.
- A possible mnemonic to remember the CIDR notation is to think of the slash as a division sign, and the number after the slash as the number of equal parts that the IP address is divided into. For example, 192.168.1.0/24 means that the IP address is divided into 24 equal parts, each with 8 bits. The first part is the network prefix, and the remaining parts are the host addresses. The subnet mask is then the number of ones in the prefix, followed by zeros in the host bits. In this case, 11111111.11111111.11111111.00000000, or 255.255.255.0.

## ARP

- ARP stands for **Address Resolution Protocol**.
- It is a communication protocol used for discovering the **link layer address**, such as a **MAC address**, associated with a given **internet layer address**, typically an **IPv4 address**.
- It is a critical function in the **Internet protocol suite**.
- It was defined in 1982 by **RFC 826**, which is **Internet Standard STD 37**.
- It works by sending **broadcast packets** to all hosts on the same network and asking for the MAC address of the host with a specific IP address.
- The host with the matching IP address replies with its MAC address, and the sender stores this information in a **cache table** for future use.
- ARP can be divided into two types: **static ARP** and **dynamic ARP**.
  - Static ARP: The sender manually enters the IP-MAC mapping in the cache table and does not need to send ARP requests.
  - Dynamic ARP: The sender dynamically learns the IP-MAC mapping by sending ARP requests and receiving ARP replies.
- ARP can also be classified into four types based on the network topology: **ARP**, **Proxy ARP**, **Reverse ARP (RARP)**, and **Inverse ARP (InARP)**.
  - ARP: The sender and the receiver are on the same network and use ARP to find each other's MAC address.
  - Proxy ARP: The sender and the receiver are on different networks and use a router as a proxy to find each other's MAC address.
  - RARP: The sender has a MAC address but no IP address and uses RARP to find its IP address from a RARP server.
  - InARP: The sender has an IP address but no MAC address and uses InARP to find its MAC address from a router.
- ARP has some advantages and disadvantages:
  - Advantages:
    - \* It is simple and easy to implement.

- \* It is flexible and adaptable to different network architectures.
- \* It reduces the network overhead by caching the IP-MAC mappings.
- Disadvantages:
  - \* It is vulnerable to **ARP spoofing** attacks, where an attacker sends fake ARP packets to trick the sender into sending data to the wrong destination.
  - \* It is inefficient and wasteful when the network size is large, as it floods the network with broadcast packets.
  - \* It does not support IPv6, which uses a different protocol called **Neighbor Discovery Protocol (NDP)**.
- A possible mnemonic to remember the four types of ARP is **PAIR**:
  - P: Proxy ARP
  - A: ARP
  - I: InARP
  - R: RARP
- A possible learning trick to understand the difference between static and dynamic ARP is to compare them to a phone book:
  - Static ARP: The sender has a phone book with the names and phone numbers of all the hosts on the network and does not need to call anyone to ask for their number.
  - Dynamic ARP: The sender does not have a phone book and has to call a directory service to ask for the phone number of a specific host.
- A possible ASCII diagram to illustrate the ARP process is:



## RARP

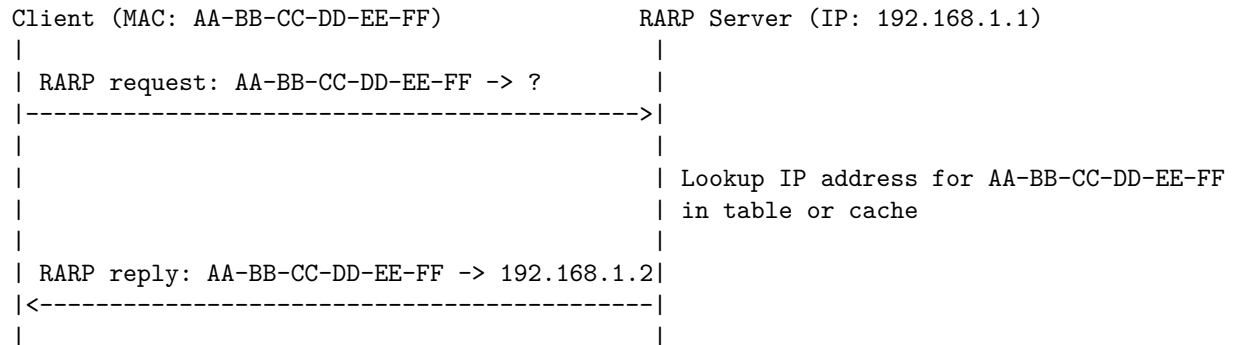
- RARP stands for **Reverse Address Resolution Protocol** .
- It is a protocol that allows a **client computer** to request its **IP address** from a **gateway server** or a **RARP server** on the same **local area network (LAN)** .
- It works by sending the **MAC address** of the client computer to the RARP server, which then looks up the corresponding IP address in a **table or cache** and sends it back to the client .
- RARP is useful for **diskless workstations** or **bootstrapping** devices that do not have a permanent IP address or a configuration file .

- RARP uses the same **packet format** as ARP, but with different **operation codes**. RARP request is 3 and RARP reply is 4 .
- RARP is an **obsolete** protocol that has been replaced by **BOOTP** and **DHCP**, which are more flexible and scalable .

A possible mnemonic to remember RARP is:

**R**everse **A**ddress **R**esolution **P**rotocol: **R**equest **A**n **R**esponse **P**lease

A possible ASCII diagram to illustrate RARP is:



## DHCP

- DHCP stands for Dynamic Host Configuration Protocol. It is a network management protocol that automatically assigns IP addresses and other communication parameters to devices connected to a network using a client-server architecture .
- DHCP enables a host to obtain an IP address dynamically when it connects to a network, without the need for manual configuration or pre-allocation. This simplifies network administration and reduces the risk of IP address conflicts .
- DHCP operates on four basic steps: discover, offer, request, and acknowledge. A DHCP client sends a broadcast message (DHCPDISCOVER) to discover a DHCP server on the network. A DHCP server responds with a unicast message (DHCPOFFER) that contains an IP address and other configuration information for the client. The client then sends another broadcast message (DHCPREQUEST) to request the offered IP address and to inform other DHCP servers that it has accepted the offer. The DHCP server confirms the IP address assignment with a unicast message (DHCPACK) to the client .
- DHCP can also provide other information to the client, such as the subnet mask, default gateway, domain name, DNS servers, and lease duration. The lease duration specifies how long the client can use the IP address

before it has to renew it or release it. DHCP can also support static IP address assignments for certain clients based on their MAC addresses .

- DHCP is defined by RFCs 2131 and 2132, and it is an extension of an earlier protocol called Bootstrap Protocol (BOOTP). DHCP supports both IPv4 and IPv6 networks, although they use different message formats and options .
- A possible mnemonic to remember the four steps of DHCP is **DORA** (Discover, Offer, Request, Acknowledge).
- A possible learning trick to understand the difference between DHCP and static IP address assignment is to compare them to renting and buying a house. DHCP is like renting a house, where you can move in quickly without much paperwork, but you have to pay rent periodically and you may have to move out if the lease expires or the landlord decides to sell the house. Static IP address assignment is like buying a house, where you have to do a lot of paperwork and pay a large amount upfront, but you have full ownership and control of the house and you can stay there as long as you want.

## ICMP

- ICMP stands for Internet Control Message Protocol. It is a network layer protocol that is used by network devices to communicate problems, errors, or diagnostics related to data transmission .
- ICMP messages are encapsulated in IP datagrams and have a specific format that consists of a type, a code, a checksum, and a data field. The type and code fields indicate the purpose and the cause of the ICMP message. The checksum field is used to verify the integrity of the message. The data field may contain additional information, such as the IP header and the first 8 bytes of the original datagram that caused the error.
- Some of the common types of ICMP messages are:
  - Echo request and echo reply: These messages are used to test the reachability and the round-trip time of a destination. They are the basis of the ping utility .
  - Destination unreachable: This message is sent by a router or a host when it cannot deliver a datagram to the intended destination. The code field specifies the reason for the failure, such as network unreachable, host unreachable, protocol unreachable, port unreachable, etc .
  - Time exceeded: This message is sent by a router or a host when the time-to-live (TTL) field of a datagram reaches zero. This prevents datagrams from looping indefinitely in the network. The code field indicates whether the TTL expired in transit or in reassembly .
  - Parameter problem: This message is sent by a router or a host when it encounters an error in the IP header of a datagram. The code



- field indicates the type of error, such as pointer indicates the error, missing a required option, bad length, etc. The data field contains a pointer to the octet where the error was detected.
- Source quench: This message is sent by a router or a host when it is experiencing congestion and cannot process or forward datagrams. The sender of the datagram should reduce its transmission rate to avoid further congestion.
  - Redirect: This message is sent by a router to inform a host of a better route to a destination. The code field specifies the type of redirect, such as redirect for network, redirect for host, redirect for type of service and network, redirect for type of service and host. The data field contains the IP address of the new router.
  - Timestamp request and timestamp reply: These messages are used to measure the time delay between two hosts. They contain a timestamp field that records the time of sending or receiving the message.
- ICMP is an important tool for network troubleshooting and management. It can help to detect network failures, misconfigurations, or performance issues. However, ICMP can also be used for malicious purposes, such as scanning, spoofing, or denial-of-service attacks. Therefore, network administrators should monitor and filter ICMP traffic carefully to ensure network security .

### **Routing in network layer**

- Routing is the process of finding the best possible path from the source host to the destination host in a network.
- Routing is performed by a special device known as a router, which works at the network layer in the OSI model and internet layer in TCP/IP model.
- A router forwards the packets based on the information available in the packet header and forwarding table, which contains the mapping of destination addresses and next-hop routers.
- Routing can be based on static tables that are rarely changed, or dynamic tables that are automatically updated depending on network conditions.
- Routing algorithms are used to determine the optimal routes for the packets. There are two main types of routing algorithms: distance vector and link state.
- Distance vector routing algorithms use the principle of “bellman-ford”, which means that each router only knows the distance and direction to its neighbors, and exchanges this information periodically with them. The distance can be measured by hop count, delay, bandwidth, or other metrics. An example of distance vector routing algorithm is RIP (Routing Information Protocol).
- Link state routing algorithms use the principle of “dijkstra”, which means

that each router knows the entire topology of the network, and calculates the shortest path to every other router using a graph algorithm. The routers exchange their link state information periodically or when there is a change in the network. An example of link state routing algorithm is OSPF (Open Shortest Path First).

- Routing can also be classified into two categories: unicast and multicast. Unicast routing is the routing of packets from one source to one destination, while multicast routing is the routing of packets from one source to multiple destinations.
- Unicast routing can be further divided into two types: intra-domain and inter-domain. Intra-domain routing is the routing of packets within a single network domain, such as a LAN or a WAN. Inter-domain routing is the routing of packets between different network domains, such as the internet. An example of intra-domain routing protocol is IGRP (Interior Gateway Routing Protocol), and an example of inter-domain routing protocol is BGP (Border Gateway Protocol).
- Multicast routing can be further divided into two types: dense mode and sparse mode. Dense mode multicast routing is the routing of packets to all routers in the network, and then pruning the branches that do not have any receivers. Sparse mode multicast routing is the routing of packets to only the routers that have receivers, and then building the branches from them. An example of dense mode multicast routing protocol is DVMRP (Distance Vector Multicast Routing Protocol), and an example of sparse mode multicast routing protocol is PIM (Protocol Independent Multicast).
- A mnemonic to remember the types of routing algorithms is “DID LID”, which stands for Distance vector, Intra-domain, Dense mode, Link state, Inter-domain, Sparse mode.

### Forwarding and Delivery in Network Layer

- The network layer supervises the handling of packets by the underlying physical networks. We call this handling as the delivery of packets to the destination.
- The delivery of a packet is called **direct** if the deliverer (host or router) and the destination are on the same network; the delivery of a packet is called **indirect** if the deliverer (host or router) and the destination are on different networks.
- The network layer also determines the route or path taken by the packets from the source to the destination. This process is called **routing**.
- Routing is a network-wide process that involves all the routers in the network. Routing algorithms are used to find the optimal paths for packet delivery.
- Forwarding is a router-local process that transfers a packet from an input link interface to the appropriate output link interface based on the routing

information.

- Forwarding requires a **routing table** that maps the destination address or network to the output link interface. The routing table is updated by the routing algorithms .
- A routing table can have **static** entries that are manually configured by the network administrator, or **dynamic** entries that are automatically updated by the routing algorithms.
- A routing table can also have **default** entries that are used when no specific match is found for the destination address or network.
- A routing table can use **address aggregation** to reduce the number of entries by combining several networks into one entry. For example, if there are four networks with addresses 200.23.16.0/24, 200.23.17.0/24, 200.23.18.0/24, and 200.23.19.0/24, they can be aggregated into one entry 200.23.16.0/22.
- A router can use different methods to find the matching entry in the routing table, such as **longest prefix matching**, **next-hop matching**, or **host-specific matching**.
- Some tools or utilities that can help with packet delivery and routing are **traceroute**, **ping**, **route**, and **ipconfig** .

: **Network Layer – Understanding Packet Delivery and Routing - Notesformsc**

Network Layer Delivery Forwarding and Routing - [PPT Powerpoint]

Data Communication and Networking - Network Layer: Delivery, Forwarding, and Routing Study Notes

Forwarding and Routing in Network Layer - Electronics Post

Network Layer Services- Packetizing, Routing and Forwarding - Geeks-forGeeks

**Static and dynamic routing in computer networks**

- Static routing and dynamic routing are two methods used to determine how to send a packet toward its destination.
- Static routes are configured in advance of any network communication by the network administrator. Dynamic routes are learned by routers through exchanging information with other routers.
- Static routing is suitable for small networks with one or two routes, while dynamic routing is more efficient for large and complex networks with multiple routes.
- Static routing provides more security and control over the network traffic, while dynamic routing adapts to network changes and failures automatically.
- Static routing uses less bandwidth and CPU resources, while dynamic routing uses more bandwidth and CPU resources to exchange routing information.
- Static routing is manual and static, while dynamic routing is automated and dynamic.

### Advantages and disadvantages of static routing

- Advantages:
  - It is simple and easy to configure.
  - It provides more security and control over the network traffic.
  - It uses less bandwidth and CPU resources than dynamic routing.
  - It is faster and more reliable than dynamic routing.
- Disadvantages:
  - It is not scalable and flexible for large and complex networks.
  - It does not adapt to network changes and failures automatically.
  - It requires manual reconfiguration of routes when the network topology changes.
  - It is prone to human errors and misconfigurations.

### Advantages and disadvantages of dynamic routing

- Advantages:
  - It is scalable and flexible for large and complex networks.
  - It adapts to network changes and failures automatically.
  - It does not require manual reconfiguration of routes when the network topology changes.
  - It can balance the network load and optimize the network performance.
- Disadvantages:
  - It is complex and difficult to configure and troubleshoot.
  - It provides less security and control over the network traffic.
  - It uses more bandwidth and CPU resources than static routing.
  - It is slower and less reliable than static routing.

### Examples of static and dynamic routing

- Static routing example:
  - Suppose there are three routers R1, R2, and R3 connected as shown below:

```
R1 ----- R2 ----- R3
| | |
| | |
A B C
```

- To configure static routes on R1, the network administrator needs to enter the following commands:  

```
R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
R1(config)# ip route 192.168.3.0 255.255.255.0 10.0.0.2
```
- These commands tell R1 to send packets destined for network 192.168.2.0/24 and 192.168.3.0/24 to the next-hop router R2 with

IP address 10.0.0.2.

- Similarly, the network administrator needs to configure static routes on R2 and R3 for the other networks.
- Dynamic routing example:
  - Suppose there are three routers R1, R2, and R3 connected as shown above, and they use the Routing Information Protocol (RIP) as the dynamic routing protocol.
  - To configure dynamic routes on R1, the network administrator needs to enter the following commands:

```
R1(config)# router rip
R1(config-router)# network 10.0.0.0
R1(config-router)# network 192.168.1.0
```

- These commands tell R1 to enable RIP on the interfaces that belong to network 10.0.0.0/24 and 192.168.1.0/24, and to advertise these networks to other routers.
- Similarly, the network administrator needs to configure dynamic routes on R2 and R3 for the other networks.
- After the configuration, the routers will exchange routing information periodically and update their routing tables accordingly.

### **Routing algorithms and protocols in computer networks**

- Routing is the process of finding and selecting the best path for data transmission from source to destination in a network.
- Routing algorithms are the software programs that implement the logic of routing, i.e., they decide the optimal path for each packet based on some criteria or metric.
- Routing protocols are the set of rules and procedures that routers use to communicate with each other and exchange routing information.
- Routing algorithms and protocols can be classified into different types based on various factors, such as:
  - The scope of the network: global or local
  - The type of information used: static or dynamic
  - The frequency of update: periodic or event-driven
  - The optimality of the path: optimal or suboptimal
  - The convergence speed: fast or slow
  - The complexity and overhead: low or high
- Some of the common routing algorithms and protocols in computer networks are :

- Distance vector routing: It is a local and dynamic routing algorithm that uses the hop count as the metric. It updates the routing table periodically by exchanging the entire table with the neighboring routers. It is simple and easy to implement, but it has drawbacks such as slow convergence, count-to-infinity problem, and routing loops. Examples of distance vector routing protocols are RIP, IGRP, and EIGRP.
  - Link state routing: It is a global and dynamic routing algorithm that uses the cost of the link as the metric. It updates the routing table only when there is a change in the network topology by flooding the link state information to all the routers. It is more efficient and accurate than distance vector routing, but it has drawbacks such as high complexity, overhead, and memory requirements. Examples of link state routing protocols are OSPF and IS-IS.
  - Path vector routing: It is a global and dynamic routing algorithm that uses the path attributes as the metric. It updates the routing table by exchanging the path information with the neighboring routers. It is more scalable and flexible than distance vector and link state routing, but it has drawbacks such as high processing and bandwidth requirements. Examples of path vector routing protocols are BGP and IDRP.
  - Hierarchical routing: It is a global and static routing algorithm that uses the network structure as the metric. It divides the network into hierarchical levels and assigns a unique address to each level. It reduces the size and complexity of the routing table by aggregating the routes, but it has drawbacks such as suboptimal paths and lack of adaptability. Examples of hierarchical routing protocols are CIDR and MPLS.
  - Broadcast routing: It is a local and static routing algorithm that uses the broadcast address as the metric. It sends the packet to all the nodes in the network or a subset of them. It is useful for applications that require multicast or group communication, but it has drawbacks such as high overhead, congestion, and security issues. Examples of broadcast routing protocols are flooding, gossiping, and spanning tree.
- Some of the mnemonics and learning tricks for routing algorithms and protocols in computer networks are:
    - To remember the difference between distance vector and link state routing, use the acronym DUAL: Distance vector Uses All Links, Link state Uses Dijkstra Algorithm.
    - To remember the difference between RIP and OSPF, use the acronym ROSE: RIP is Old, Slow, and Easy, OSPF is Recent, Optimal, and Secure.
    - To remember the difference between BGP and MPLS, use the acronym BMP: BGP is for Border Gateway Protocol, MPLS is for Multi-Protocol Label Switching.

## Congestion control algorithms in computer networks

- Congestion control algorithms are mechanisms that control the entry of data packets into the network, enabling a better use of a shared network infrastructure and avoiding congestive collapse.
- Congestive collapse is a situation where the network performance degrades drastically due to excessive traffic and congestion.
- Congestion control algorithms can be broadly classified into two categories: open loop and closed loop.
- Open loop congestion control policies are applied to prevent congestion before it happens. They involve designing the network and choosing the appropriate protocols and parameters to avoid congestion.
- Closed loop congestion control policies are applied to detect and mitigate congestion after it happens. They involve monitoring the network state and adjusting the transmission rate or window size of the senders based on feedback signals such as packet loss, delay, or explicit notifications.
- Some examples of open loop congestion control techniques are admission control, traffic shaping, and resource reservation.
- Some examples of closed loop congestion control techniques are congestion avoidance, congestion recovery, and congestion notification.
- Congestion avoidance algorithms are implemented at the TCP layer as the mechanism to avoid congestive collapse in a network. They aim to keep the network operating at the optimal point where the throughput is high and the delay is low.
- Some examples of congestion avoidance algorithms are slow start, additive increase/multiplicative decrease (AIMD), congestion window (CWND), and fast retransmit.
- Slow start is an algorithm that starts with a small transmission rate and increases it exponentially until a threshold is reached or a packet is lost. It is used to probe the network capacity and avoid sending too many packets at once.
- AIMD is an algorithm that increases the transmission rate linearly until a packet is lost, and then decreases it multiplicatively by a factor. It is used to achieve fairness and stability among multiple competing flows.
- CWND is a variable that limits the number of packets that a sender can have in the network at any time. It is adjusted by the congestion avoidance algorithms based on the network conditions.
- Fast retransmit is an algorithm that detects packet loss by receiving duplicate acknowledgments from the receiver, and retransmits the lost packet without waiting for a timeout. It is used to recover from packet loss quickly and avoid unnecessary retransmissions.
- Some mnemonics and learning tricks for congestion control algorithms are:
  - Slow start: Start slow and grow fast.
  - AIMD: Add and multiply, divide and subtract.
  - CWND: Window size is the key to congestion control.
  - Fast retransmit: Don't wait, just retransmit.

## IPv6 in cn

- IPv6 is the latest version of the Internet Protocol (IP), which is the set of rules that defines how devices communicate over the Internet.
- IPv6 was designed to overcome the limitations of IPv4, such as the exhaustion of address space, the lack of end-to-end connectivity, and the complexity of network configuration and management.
- IPv6 uses 128-bit addresses, which can support  $2^{128}$  (approximately  $3.4 \times 10^{38}$ ) possible addresses, compared to  $2^{32}$  (approximately 4.3 billion) addresses in IPv4.
- IPv6 also introduces new features, such as stateless address autoconfiguration, neighbor discovery, multicast, anycast, mobility, security, and transition mechanisms.
- China is one of the leading countries in IPv6 deployment and adoption, with the goal of running a single-stack IPv6 network by 2030, and having 700 million active IPv6 users and 200 million Internet of Things devices using IPv6 by 2023.
- China has allocated a total of 1,024 /32 IPv6 prefixes to various entities, such as Internet service providers, universities, enterprises, and government agencies, as of December 2018.
- China has also implemented several IPv6 projects and initiatives, such as the China Next Generation Internet (CNGI), the China Education and Research Network (CERNET), the China IPv6 Council, and the China IPv6 Forum, to promote the research, development, and application of IPv6 in the country.
- Some of the benefits of IPv6 for China include:
  - Enhancing the network performance, security, and scalability for the growing number of Internet users and devices in China.
  - Supporting the development of emerging technologies and applications, such as cloud computing, big data, artificial intelligence, and 5G.
  - Improving the global connectivity and interoperability of China's Internet infrastructure and services.
  - Contributing to the innovation and leadership of China in the Internet industry and governance.
- Some of the challenges of IPv6 for China include:
  - Coordinating the transition and coexistence of IPv4 and IPv6 among different stakeholders and sectors.
  - Addressing the technical, economic, and policy issues related to IPv6 deployment and adoption.
  - Enhancing the awareness, education, and training of IPv6 among the public and professionals.
  - Developing and testing the IPv6 compatibility and functionality of various devices, software, and applications.
- Some of the mnemonics and learning tricks for IPv6 in cn include:
  - Remembering the IPv6 address format: 8 groups of 4 hexadecimal



digits, separated by colons, with leading zeros and consecutive zeros omitted. For example, 2001:0db8:0000:0000:ff00:0042:8329 can be written as 2001:db8::ff00:42:8329.

- Remembering the IPv6 address types: unicast (one-to-one), multicast (one-to-many), anycast (one-to-nearest), and broadcast (one-to-all). Unicast addresses start with 2 or 3, multicast addresses start with ff, anycast addresses are a subset of unicast addresses, and broadcast addresses are replaced by multicast addresses in IPv6.
- Remembering the IPv6 address scopes: global (worldwide), unique local (private), link-local (local network), and site-local (deprecated). Global addresses start with 2 or 3, unique local addresses start with fc or fd, link-local addresses start with fe80, and site-local addresses start with fec0.
- Remembering the IPv6 special addresses: loopback (::1), unspecified (::), IPv4-mapped (::ffff:0:0/96), and IPv4-compatible (::/96, deprecated). Loopback address is used to test the local host, unspecified address is used when the source address is unknown, IPv4-mapped address is used to represent an IPv4 address in an IPv6 context, and IPv4-compatible address is used to tunnel IPv6 packets over IPv4 networks.

## Unit 4 - Transport Layer in Computer Networks

- The transport layer is a conceptual division of methods in the layered architecture of protocols in the network stack in the Internet protocol suite and the OSI model.
- The protocols of this layer provide end-to-end communication services for applications.
- The transport layer takes data from the application layer and then breaks it into smaller size segments, numbers each byte, and hands over to the network layer for delivery.
- The transport layer also reassembles the segments at the destination and delivers the complete message to the application layer.
- The transport layer performs the following functions :
  - **Addressing:** The transport layer provides the user address which is specified as a station or port. The port variable represents a specific process or service running on a host. The transport layer protocols need to know the port number of the source and destination processes to deliver the data correctly.
  - **Multiplexing and Demultiplexing:** The transport layer can multiplex multiple application processes on a single host by assigning them different port numbers and sending their data segments over the same network layer connection. The transport layer can also demultiplex the incoming data segments by using the port numbers to deliver them to the correct application processes.
  - **Reliability:** The transport layer can provide reliable or unreliable

delivery of data depending on the protocol used. Reliable delivery means that the transport layer ensures that the data segments are delivered to the destination without errors, loss, duplication, or out-of-order arrival. Unreliable delivery means that the transport layer does not guarantee the delivery of data segments and leaves the error detection and correction to the application layer.

- **Flow Control:** The transport layer can regulate the rate of data transmission between the source and destination hosts to avoid congestion and buffer overflow. The transport layer can use various mechanisms such as sliding window, stop-and-wait, or feedback-based to control the flow of data segments.
- **Error Control:** The transport layer can detect and correct errors in the data segments by using techniques such as checksum, acknowledgment, retransmission, or timeout. The transport layer can also handle the corrupted, lost, or duplicated segments by discarding them or requesting for retransmission.
- **Congestion Control:** The transport layer can prevent or reduce the congestion in the network by adjusting the rate of data transmission according to the network conditions. The transport layer can use various algorithms such as additive increase multiplicative decrease (AIMD), slow start, congestion avoidance, or congestion recovery to control the congestion.
- **Quality of Service:** The transport layer can provide different levels of quality of service (QoS) to different applications based on their requirements. The transport layer can use parameters such as bandwidth, delay, jitter, or loss to measure and guarantee the QoS.
- The transport layer can use different protocols to provide different services to the applications. Some of the common transport layer protocols are :
  - **Transmission Control Protocol (TCP):** TCP is a connection-oriented, reliable, byte-stream protocol that provides error control, flow control, and congestion control. TCP is widely used for applications that require reliable and ordered delivery of data, such as web browsing, email, file transfer, or remote login .
  - **User Datagram Protocol (UDP):** UDP is a connectionless, unreliable, datagram protocol that provides minimal error control and no flow control or congestion control. UDP is widely used for applications that require fast and efficient delivery of data, such as voice over IP (VoIP), video streaming, online gaming, or domain name system (DNS) .
  - **Stream Control Transmission Protocol (SCTP):** SCTP is a connection-oriented, reliable, message-oriented protocol that provides error control, flow control, congestion control, and multiplexing. SCTP is widely used for applications that require multiple streams of data, such as telephony signaling, web conferencing, or multimedia transmission .
  - **Datagram Congestion Control Protocol (DCCP):** DCCP is

a connection-oriented, unreliable, datagram protocol that provides congestion control and QoS. DCCP is widely used for applications that require controlled rate of data transmission,

### **Process-to-process delivery in transport layer**

- The transport layer is responsible for delivering data from one process to another process on different hosts.
- A process is an entity of the application layer that uses the services of the transport layer.
- The transport layer uses the host-to-host delivery service of the network layer to send data between hosts, and then adds its own functionality to enable process-to-process delivery.
- The transport layer uses two methods to identify the processes on the source and destination hosts: port numbers and socket addresses.

### **Port numbers**

- A port number is a 16-bit integer that identifies a specific process on a host.
- The port numbers are divided into three ranges: well-known ports (0 to 1023), registered ports (1024 to 49151), and dynamic or private ports (49152 to 65535).
- Well-known ports are assigned by the Internet Assigned Numbers Authority (IANA) to standard services, such as HTTP (80), FTP (21), and Telnet (23).
- Registered ports are assigned by IANA to user applications, such as email clients, web browsers, and chat applications.
- Dynamic or private ports are used by processes that need temporary communication, such as client processes that initiate a connection to a server.

### **Socket addresses**

- A socket address is a combination of an IP address and a port number that uniquely identifies a process on a host.
- A socket address is written as IP address:port number, such as 192.168.1.1:80.
- The transport layer uses the socket address to deliver data to the correct process on the destination host.

### **Process-to-process delivery example**

- Suppose a web browser on host A wants to request a web page from a web server on host B.
- The web browser is a client process that uses a dynamic port number, such as 5000, to communicate with the web server.

- The web server is a server process that uses a well-known port number, such as 80, to listen for incoming requests.
- The web browser creates a socket address by combining its own IP address and port number, such as 192.168.1.1:5000.
- The web browser also creates a socket address for the web server by combining its IP address and port number, such as 192.168.1.2:80.
- The web browser sends a request to the web server using the socket addresses as the source and destination addresses.
- The transport layer on host A adds the socket addresses to the data and passes it to the network layer.
- The network layer on host A adds the IP addresses to the data and passes it to the data link layer.
- The data link layer on host A adds the MAC addresses to the data and sends it over the link to host B.
- The data link layer on host B removes the MAC addresses and passes the data to the network layer.
- The network layer on host B removes the IP addresses and passes the data to the transport layer.
- The transport layer on host B checks the destination socket address and delivers the data to the web server process on port 80.
- The web server process sends a response to the web browser using the same socket addresses as the source and destination addresses.
- The transport layer on host B adds the socket addresses to the data and passes it to the network layer.
- The network layer on host B adds the IP addresses to the data and passes it to the data link layer.
- The data link layer on host B adds the MAC addresses to the data and sends it over the link to host A.
- The data link layer on host A removes the MAC addresses and passes the data to the network layer.
- The network layer on host A removes the IP addresses and passes the data to the transport layer.
- The transport layer on host A checks the destination socket address and delivers the data to the web browser process on port 5000.

#### ASCII diagram of process-to-process delivery

|                  |                 |
|------------------|-----------------|
| +-----+          | +-----+         |
| Web browser      | Web server      |
| Port: 5000       | Port: 80        |
| +-----+          | +-----+         |
| Transport layer  | Transport layer |
| Socket address   | Socket address  |
| 192.168.1.1:5000 | 192.168.1.2:80  |
| +-----+          | +-----+         |
| Network layer    | Network layer   |

|             |  |            |  |
|-------------|--|------------|--|
| IP address  |  | IP address |  |
| 192.168.1.1 |  | 192.168.1  |  |

## Transport layer protocols

- Transport layer protocols are responsible for providing end-to-end communication between applications on different hosts in a network.
- Transport layer protocols can offer various services such as reliability, flow control, congestion control, multiplexing, error detection, and security.
- The two most common transport layer protocols are Transmission Control Protocol (TCP) and User Datagram Protocol (UDP).

## TCP

- TCP is a connection-oriented protocol that establishes a logical connection between the sender and the receiver before exchanging data.
- TCP provides reliable data transfer by using sequence numbers, acknowledgments, timers, and retransmission mechanisms to ensure that all data segments are delivered correctly and in order.
- TCP also provides flow control by using a sliding window mechanism to regulate the amount of data that can be sent by the sender at a time, based on the receiver's buffer capacity and network conditions.
- TCP also provides congestion control by using various algorithms to adjust the sender's window size and transmission rate according to the network congestion level, to avoid overwhelming the network and causing packet loss.
- TCP uses a three-way handshake to establish a connection, and a four-way handshake to terminate a connection.
- TCP uses port numbers to identify different applications on the same host, and to multiplex and demultiplex data streams.
- TCP is suitable for applications that require reliable and ordered delivery of data, such as web browsing, email, file transfer, and remote login.

## UDP

- UDP is a connectionless protocol that does not establish a logical connection between the sender and the receiver before exchanging data.
- UDP provides unreliable data transfer by sending data segments without any guarantee of delivery, order, or error detection.
- UDP does not provide any flow control or congestion control mechanisms, and relies on the application layer to handle these issues.
- UDP does not use any handshaking or connection management procedures, and thus has less overhead and latency than TCP.
- UDP also uses port numbers to identify different applications on the same host, and to multiplex and demultiplex data streams.
- UDP is suitable for applications that require fast and real-time delivery of data, such as voice over IP, video streaming, online gaming, and DNS.

## Comparison of TCP and UDP

| Feature            | TCP                                     | UDP                       |
|--------------------|-----------------------------------------|---------------------------|
| Connection         | Connection-oriented                     | Connectionless            |
| Reliability        | Reliable                                | Unreliable                |
| Ordering           | Ordered                                 | Unordered                 |
| Error detection    | Yes                                     | No                        |
| Flow control       | Yes                                     | No                        |
| Congestion control | Yes                                     | No                        |
| Handshaking        | Yes                                     | No                        |
| Overhead           | High                                    | Low                       |
| Latency            | High                                    | Low                       |
| Port numbers       | Yes                                     | Yes                       |
| Applications       | Web, email, file transfer, remote login | Voice, video, gaming, DNS |

## Mnemonics and learning tricks

- To remember the features of TCP and UDP, you can use the following acronyms:
  - TCP: **T**ransport **C**onnection **P**rotocol
  - UDP: **U**nreliable **D**atagram **P**rotocol
- To remember the difference between connection-oriented and connectionless protocols, you can use the following analogy:
  - Connection-oriented protocols are like making a phone call, where you have to dial a number, wait for the other person to answer, and then talk. You also have to say goodbye and hang up when you are done.
  - Connectionless protocols are like sending a postcard, where you just write a message, put a stamp, and drop it in the mailbox. You don't know if the other person will receive it, read it, or reply to it.
- To remember the difference between reliable and unreliable protocols, you can use the following analogy:
  - Reliable protocols are like sending a registered mail, where you get a receipt, a tracking number, and a confirmation of delivery. You can also resend the mail if it gets lost or damaged.
  - Unreliable protocols are like sending a regular mail, where you don't get any receipt, tracking number, or confirmation of delivery. You also don't know if the mail gets lost, damaged, or delivered out of order.

## UDP Transport layer protocol

- UDP stands for User Datagram Protocol. It is a transport layer protocol that is part of the Internet Protocol suite, also known as UDP/IP suite.
- UDP is a simple, unreliable and connectionless protocol. It does not establish a connection before sending data, nor does it guarantee that the data will be delivered or received in order .
- UDP adds a 16-byte header to the data from the upper layer. The header contains four fields: source port, destination port, length and checksum. The source and destination ports identify the endpoints of the communication. The length field specifies the total length of the UDP datagram, including the header and the data. The checksum field provides error detection for the header and the data .

| Source Port | Destination Port |
|-------------|------------------|
| 16 bits     | 16 bits          |

| Length  | Checksum |
|---------|----------|
| 16 bits | 16 bits  |

- UDP is suitable for applications that require fast, low-overhead and one-way communication, such as streaming audio and video, online gaming, DNS queries and SNMP messages .
- UDP has some advantages and disadvantages over TCP, the other major transport layer protocol. Some of the advantages are:
  - UDP is simpler and more efficient than TCP, as it does not involve connection setup, acknowledgment, retransmission, congestion control or flow control .
  - UDP allows multicast and broadcast communication, as it can send data to multiple destinations with a single send operation .
  - UDP gives more control to the application layer, as it can decide how to handle the reliability, ordering and congestion issues .
- Some of the disadvantages are:
  - UDP does not provide reliability, ordering or congestion control, which may result in data loss, duplication, delay or jitter .
  - UDP does not provide any mechanism for flow control, which may cause the sender to overwhelm the receiver or the network .
  - UDP is vulnerable to spoofing attacks, as it does not verify the source of the datagrams.
- A possible mnemonic to remember the four fields of the UDP header is: **S**ome **D**ogs **L**ove **C**heese, which stands for **S**ource port, **D**estination port, **L**ength and **C**hecksum. Alternatively, one can use the acronym **SDLC** and associate it with something memorable, such as **S**uper **D**uper **L**ovely

Cat.

### **TCP Transport layer protocol**

- TCP stands for Transmission Control Protocol. It is one of the core protocols of the Internet Protocol Suite, which provides reliable, ordered, and error-checked delivery of data between applications running on hosts connected by a network.
- TCP operates at the transport layer, which is the fourth layer of the OSI model. The transport layer is responsible for providing end-to-end communication services for applications, such as multiplexing, segmentation, flow control, congestion control, and error detection and correction.
- TCP uses a connection-oriented approach, which means that before any data can be exchanged, a TCP connection must be established between the sender and the receiver. A TCP connection is identified by a four-tuple: source IP address, source port number, destination IP address, and destination port number. A port number is a 16-bit number that identifies a specific process or application running on a host.
- TCP uses a three-way handshake to establish a connection. The three steps are:
  - The sender sends a SYN (synchronize) segment to the receiver, which contains the sender's initial sequence number and other parameters, such as window size and maximum segment size.
  - The receiver replies with a SYN-ACK (synchronize-acknowledge) segment, which contains the receiver's initial sequence number and an acknowledgment of the sender's sequence number, as well as other parameters.
  - The sender responds with an ACK (acknowledge) segment, which acknowledges the receiver's sequence number and completes the connection establishment.
- TCP uses a sliding window mechanism to control the flow of data and to ensure reliable delivery. A sliding window is a variable-sized buffer that indicates how much data can be sent or received at any given time. The sender maintains a send window, which indicates how much data it can send before receiving an acknowledgment from the receiver. The receiver maintains a receive window, which indicates how much data it can receive before sending an acknowledgment to the sender. The size of the window can vary depending on the network conditions and the availability of buffer space at both ends.
- TCP uses sequence numbers and acknowledgments to keep track of the data segments that are sent and received. A sequence number is a 32-bit number that identifies the position of a byte in the data stream. An



acknowledgment is a 32-bit number that indicates the next expected sequence number from the sender. TCP uses cumulative acknowledgments, which means that an acknowledgment of a sequence number implies that all the previous segments have been received correctly.

- TCP uses a checksum to detect errors in the data segments. A checksum is a 16-bit value that is computed by adding up all the 16-bit words in the segment, including the header and the payload, and taking the one's complement of the result. The checksum is stored in the header of the segment and is verified by the receiver. If the checksum does not match, the segment is discarded and an acknowledgment is not sent.
- TCP uses timers and retransmission to handle lost or corrupted segments. A timer is a value that indicates how long the sender should wait for an acknowledgment before retransmitting a segment. A retransmission is the process of sending a segment again after a timeout or a duplicate acknowledgment. TCP uses various algorithms to adjust the timer and the retransmission strategy, such as exponential backoff, fast retransmit, and fast recovery.
- TCP uses congestion control to avoid overloading the network with too much data. Congestion control is the process of adjusting the sending rate of the sender based on the feedback from the network. TCP uses various algorithms to implement congestion control, such as slow start, congestion avoidance, and congestion recovery. These algorithms use parameters such as congestion window, threshold, and duplicate acknowledgments to determine the optimal sending rate.
- TCP uses flags to indicate the state of the connection and to signal the termination of the connection. A flag is a 1-bit value that is set or cleared in the header of the segment. TCP uses six flags: SYN, ACK, FIN, RST, PSH, and URG. The meanings of these flags are:
  - SYN: used to initiate a connection
  - ACK: used to acknowledge a segment
  - FIN: used to indicate the end of data transmission
  - RST: used to reset a connection
  - PSH: used to indicate that the data should be pushed to the application immediately
  - URG: used to indicate that the data contains urgent information
- TCP uses a four-way handshake to terminate a connection. The four steps are:
  - The sender sends a FIN segment to the receiver, which indicates that it has no more data to send.
  - The receiver replies with an ACK segment, which acknowledges the sender's FIN segment.
  - The receiver sends a FIN segment to the sender

## Multiplexing in transport layer

- Multiplexing is the process of collecting the data from multiple application processes of the sender, enveloping that data with headers and sending them as a whole to the intended receiver .
- Multiplexing extends the host-to-host delivery service provided by the network layer to a process-to-process delivery service for applications running on the hosts .
- Multiplexing takes a parallelized source of information and turns it into a serialized format.
- Multiplexing is done by adding a transport layer header to each data segment, which contains the source and destination port numbers that identify the corresponding application processes .
- The port numbers are 16-bit integers that range from 0 to 65535. Some port numbers are reserved for well-known services, such as HTTP (80), FTP (21), SMTP (25), etc .
- The transport layer header also contains other fields, such as sequence number, acknowledgment number, checksum, etc., depending on the protocol used (TCP or UDP) .
- The transport layer header and the data segment together form a transport layer segment, which is then passed to the network layer for further processing .
- An example of multiplexing in transport layer is shown below:

```
Application layer data from process A: Hello
Application layer data from process B: World
Transport layer header for process A: Source port = 1000, Destination port = 2000
Transport layer header for process B: Source port = 1001, Destination port = 2001
Transport layer segment for process A: 1000 2000 Hello
Transport layer segment for process B: 1001 2001 World
```

- A mnemonic to remember the concept of multiplexing in transport layer is: **Many Processes Send Data Segments (MPSDS)**.
- A learning trick to understand the concept of multiplexing in transport layer is to imagine a post office that collects letters from different senders, adds the addresses of the recipients, and sends them as a single package to another post office.

## Connection management in transport layer

- Connection management is the process of establishing, maintaining, and terminating a logical connection between two or more hosts in a network.
- Connection management is performed by the transport layer protocols, such as TCP and UDP, to provide reliable or unreliable end-to-end communication services to the application layer.
- Connection management involves three phases: connection establishment,

data transfer, and connection termination.

- Connection establishment is the phase where the hosts agree on the parameters and rules for the data transfer, such as the port numbers, sequence numbers, window sizes, etc. Connection establishment can be active or passive, depending on whether the host initiates or waits for a connection request.
- Data transfer is the phase where the hosts exchange data packets according to the agreed parameters and rules. Data transfer can be reliable or unreliable, depending on whether the transport layer protocol guarantees the delivery, order, and integrity of the data packets. Reliable data transfer protocols, such as TCP, use mechanisms such as acknowledgments, retransmissions, flow control, and congestion control to ensure the quality of service. Unreliable data transfer protocols, such as UDP, do not use such mechanisms and leave the error handling and recovery to the application layer.
- Connection termination is the phase where the hosts gracefully or abruptly end the logical connection and release the resources allocated for the data transfer. Connection termination can be initiated by either host or by a network failure. Connection termination can be symmetric or asymmetric, depending on whether the hosts exchange messages to confirm the end of the connection or not.
- A mnemonic to remember the three phases of connection management is: **E**stablish, **T**ransfer, **T**erminate (**ETT**).
- A learning trick to understand the difference between reliable and unreliable data transfer protocols is to compare them with postal services. Reliable data transfer protocols are like registered mail, where the sender gets a receipt when the mail is delivered, and can track the status of the mail along the way. Unreliable data transfer protocols are like ordinary mail, where the sender does not get any confirmation or feedback about the delivery of the mail, and has no way of knowing if the mail was lost, delayed, or damaged.

### Flow control in transport layer

- Flow control is a mechanism that regulates the rate of data transmission between a sender and a receiver.
- Flow control ensures that the sender does not overwhelm the receiver with more data than it can process or store.
- Flow control is implemented by using feedback messages from the receiver to the sender, indicating how much data the receiver can accept at a given time.
- Flow control can be classified into two types: stop-and-wait and sliding window.

### **Stop-and-wait flow control**

- In stop-and-wait flow control, the sender sends one data unit (such as a packet or a frame) and waits for an acknowledgment from the receiver before sending the next data unit.
- The sender can only have one unacknowledged data unit in transit at any time.
- The receiver sends an acknowledgment after receiving and processing each data unit.
- The acknowledgment can also include a request to stop sending data if the receiver's buffer is full or a request to resume sending data if the receiver's buffer has space.
- Stop-and-wait flow control is simple and reliable, but it has low efficiency and high latency, especially for long-distance or noisy channels.

### **Sliding window flow control**

- In sliding window flow control, the sender can send multiple data units without waiting for acknowledgments, as long as the number of unacknowledged data units does not exceed a certain limit, called the window size.
- The window size is determined by the receiver's buffer capacity and the channel's bandwidth-delay product (the product of the channel's bandwidth and the round-trip time between the sender and the receiver).
- The sender maintains a sending window, which is a range of sequence numbers of the data units that can be sent or have been sent but not yet acknowledged.
- The receiver maintains a receiving window, which is a range of sequence numbers of the data units that can be accepted or have been accepted but not yet delivered to the application layer.
- The receiver sends acknowledgments for the data units it receives, and also indicates the current window size or the next expected sequence number.
- The sender slides its window forward when it receives an acknowledgment, and the receiver slides its window forward when it delivers a data unit to the application layer.
- Sliding window flow control can be further divided into two subtypes: go-back-N and selective repeat.

### **Go-back-N sliding window flow control**

- In go-back-N sliding window flow control, the sender can send up to N data units without waiting for acknowledgments, where N is the window size.
- The receiver only sends cumulative acknowledgments, which acknowledge all the data units up to a certain sequence number.
- The receiver only accepts data units in order, and discards any out-of-order data units.

- If the sender does not receive an acknowledgment for a data unit within a certain time, it assumes that the data unit or its acknowledgment was lost, and retransmits all the data units starting from that sequence number.
- Go-back-N sliding window flow control is simple and easy to implement, but it has low efficiency and high overhead, especially for large window sizes or high error rates.

### **Selective repeat sliding window flow control**

- In selective repeat sliding window flow control, the sender can send up to N data units without waiting for acknowledgments, where N is the window size.
- The receiver sends individual acknowledgments for each data unit it receives, and also indicates the current window size or the next expected sequence number.
- The receiver can accept data units out of order, and buffers them until they can be delivered in order to the application layer.
- If the sender does not receive an acknowledgment for a data unit within a certain time, it assumes that the data unit or its acknowledgment was lost, and retransmits only that data unit.
- Selective repeat sliding window flow control is more efficient and less wasteful than go-back-N sliding window flow control, but it is more complex and requires more buffer space at the receiver.

### **etransmission in transport layer**

- The transport layer is the fourth layer of the OSI model and the third layer of the TCP/IP model. It provides end-to-end communication services for applications, such as reliable data transfer, flow control, congestion control, error detection and correction, and multiplexing.
- etransmission is a term that refers to the process of sending and receiving data packets over a network using the transport layer protocols, such as TCP and UDP .
- etransmission involves the following steps:
  - Segmentation: The transport layer divides the data stream from the application layer into smaller units called segments, which have a header and a payload. The header contains information such as the source and destination port numbers, sequence numbers, and checksums.
  - Multiplexing: The transport layer assigns a unique port number to each application process that communicates over the network, and uses the port numbers to identify the segments belonging to different processes. This allows multiple processes to share the same network connection.
  - Error detection and correction: The transport layer uses checksums to verify the integrity of the segments, and sends acknowledgments

- (ACKs) or negative acknowledgements (NACKs) to inform the sender about the status of the data transmission. If a segment is lost, corrupted, or duplicated, the transport layer can request a retransmission of the segment using the sequence numbers .
- Flow control: The transport layer regulates the rate of data transmission between the sender and the receiver, to avoid overwhelming the receiver or the network. The transport layer uses mechanisms such as sliding window, stop-and-wait, and backpressure to control the amount of data that can be sent or received at a time .
- Congestion control: The transport layer monitors the network conditions and adjusts the data transmission rate accordingly, to avoid or reduce congestion in the network. The transport layer uses mechanisms such as additive increase multiplicative decrease (AIMD), slow start, congestion avoidance, and fast recovery to control the congestion .
- etransmission can be performed using different transport layer protocols, such as TCP and UDP. TCP is a connection-oriented, reliable, and full-duplex protocol that provides all the features of etransmission, such as segmentation, multiplexing, error detection and correction, flow control, and congestion control. UDP is a connectionless, unreliable, and simplex protocol that provides only segmentation and multiplexing, and does not guarantee the delivery, order, or integrity of the segments .
- etransmission is important for ensuring the quality of service (QoS) and the performance of the network applications, such as web browsing, email, file transfer, video streaming, online gaming, and voice over IP (VoIP). etransmission enables the applications to communicate effectively and efficiently over the network, without worrying about the details of the lower layers.

### Window management in transport layer

- Window management is a technique used by the transport layer protocols, such as TCP, to control the flow of data between two end systems.
- Window management ensures that the sender does not overwhelm the receiver or the network with too many packets at once, and that the receiver can acknowledge the received packets in a timely manner.
- Window management uses a sliding window protocol, which is a variable-sized buffer that moves along the sequence of packets. The window size indicates how many packets can be sent or received at a time, without waiting for an acknowledgment.
- The sender maintains a send window, which is the range of packets that it can send before receiving an acknowledgment from the receiver. The receiver maintains a receive window, which is the range of packets that it can receive and buffer before sending an acknowledgment to the sender.
- The sender and the receiver exchange window size information in the header of each packet, and adjust their windows accordingly. The window

size can vary depending on the network conditions, such as congestion, delay, or errors.

- The sliding window protocol can be either stop-and-wait or go-back-N. In stop-and-wait, the sender sends one packet at a time and waits for an acknowledgment before sending the next packet. The window size is one in this case. In go-back-N, the sender can send up to N packets without waiting for an acknowledgment, where N is the window size. If an acknowledgment is not received within a certain time, the sender retransmits all the packets in the window.
- A mnemonic to remember the difference between stop-and-wait and go-back-N is: Stop-and-wait is slow, go-back-N is fast.

### TCP Congestion Control in Transport Layer

- TCP stands for Transmission Control Protocol. It is a transport layer protocol that facilitates the transmission of packets from source to destination. It is a connection-oriented protocol that means it establishes the connection prior to the communication that occurs between the computing devices in a network.
- TCP is a reliable protocol as it follows the flow and error control mechanism. Flow control ensures that the sender does not overwhelm the receiver with more data than it can handle. Error control detects and corrects the errors that may occur during the transmission of packets.
- Congestion control is another mechanism that TCP uses to avoid or reduce the congestion in the network. Congestion occurs when the network resources, such as routers, links, or buffers, are overloaded and cannot handle the incoming traffic. Congestion can cause packet loss, delay, and reduced throughput.
- TCP congestion control aims to achieve two goals: efficiency and fairness. Efficiency means that the network utilization is maximized and the packet loss is minimized. Fairness means that each TCP connection gets an equal share of the network bandwidth.
- There are three phases that TCP uses for congestion control: slow start, congestion avoidance, and congestion detection:
  - Slow start: In the first phase of congestion control, the sender starts with a small number of packets, called the congestion window (cwnd), and gradually increases it by one packet for each acknowledgment (ACK) received from the receiver. This way, the sender probes the network capacity and avoids sending too many packets at once. The slow start phase ends when the cwnd reaches a threshold value (ssthresh), which is dynamically adjusted based on the network conditions.
  - Congestion avoidance: In the second phase of congestion control, the sender increases the cwnd more slowly, by one packet per round-trip time (RTT), instead of one packet per ACK. This way, the sender tries to maintain a steady flow of packets and avoid causing congestion.

tion. The congestion avoidance phase continues until a packet loss is detected, which indicates that the network is congested.

- Congestion detection: In the third phase of congestion control, the sender detects the congestion by either a timeout or a duplicate ACK. A timeout occurs when the sender does not receive an ACK for a packet within a certain time interval. A duplicate ACK occurs when the receiver sends an ACK for a packet that it has already acknowledged, which implies that some packets are missing or out of order. When a congestion is detected, the sender reduces the cwnd by half and sets the ssthresh to the new cwnd value. Then, the sender enters the slow start phase again and repeats the process.
- TCP congestion control uses an algorithm called AIMD (Additive Increase Multiplicative Decrease) to adjust the cwnd. AIMD increases the cwnd additively in the congestion avoidance phase and decreases it multiplicatively in the congestion detection phase. This algorithm ensures that the cwnd oscillates around the optimal value and converges to a fair allocation of the network bandwidth among the TCP connections.
- A mnemonic to remember the three phases of TCP congestion control is SAC: Slow start, Avoidance, and Congestion detection. A mnemonic to remember the AIMD algorithm is AMID: Additive, Multiplicative, Increase, and Decrease.

### Quality of service in transport layer

- Quality of service (QoS) is the use of mechanisms or technologies that work on a network to control traffic and ensure the performance of critical applications with limited network capacity.
- The transport layer is responsible for enhancing the QoS provided by the network layer by offering reliable and efficient end-to-end data delivery between application processes.
- The transport layer can provide QoS by using the following methods:
  - Establishing transport connections by sending requests and specifying the required QoS parameters, such as throughput, delay, jitter, and reliability.
  - Using transport addresses to identify the end points of a connection, such as port numbers in the internet.
  - Providing flow control and congestion control to regulate the rate of data transmission and avoid network overload.
  - Providing error control and retransmission to ensure data integrity and reliability.
  - Providing multiplexing and demultiplexing to allow multiple applications to share the same network resources.
  - Providing segmentation and reassembly to divide large data units into smaller ones for efficient transmission and recombine them at the destination.
- The transport layer can use different protocols to provide different levels



of QoS, such as TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) in the internet.

- TCP is a connection-oriented, reliable, and full-duplex protocol that provides flow control, congestion control, error control, and retransmission.
- UDP is a connectionless, unreliable, and simplex protocol that provides minimal QoS and is suitable for real-time applications that can tolerate some data loss.
- A mnemonic to remember the main functions of the transport layer is **FECMSS** (Flow control, Error control, Congestion control, Multiplexing, Segmentation, and Synchronization).

## Unit 5 - Application Layer in Computer Networks

- The application layer is the highest layer in the protocol suite. It provides services to the user and interacts with network applications .
- Examples of network applications are web browsers, chat clients, email clients, file transfer, remote login, etc .
- The application layer uses a logical connection, which means that the two application layers assume that there is an imaginary direct connection between them.
- The application layer can use different paradigms for communication, such as client-server, peer-to-peer, or hybrid.
- Client-server paradigm: The service provider is an application program called the server process, and the service requester is another application program called the client process. The server is usually passive and waits for requests from clients, while the client is usually active and initiates the communication. Examples of client-server applications are web, email, FTP, etc.
- Peer-to-peer paradigm: The service provider and the service requester are both application programs running on different peers, which are computers that are not dedicated to be either a client or a server. The peers communicate directly with each other without using a dedicated server. Examples of peer-to-peer applications are file sharing, voice over IP, etc.
- Hybrid paradigm: The service provider and the service requester are both application programs running on different peers, but they also use some servers for coordination or indexing purposes. Examples of hybrid applications are BitTorrent, Skype, etc.
- The application layer protocols define the rules and formats for exchanging messages between applications . They also specify the syntax and semantics of the messages, such as the type, length, order, and meaning of the fields.
- Some common application layer protocols are:

| Protocol | Description                                                                                                   | Port number              |
|----------|---------------------------------------------------------------------------------------------------------------|--------------------------|
| HTTP     | Hypertext Transfer Protocol. It is used to transfer web pages from a web server to a web browser.             | 80                       |
| HTTPS    | Hypertext Transfer Protocol Secure. It is a secure version of HTTP that uses encryption and authentication.   | 443                      |
| FTP      | File Transfer Protocol. It is used to transfer files between a client and a server.                           | 20 (data), 21 (control)  |
| SMTP     | Simple Mail Transfer Protocol. It is used to send email messages from a sender to a receiver.                 | 25                       |
| POP3     | Post Office Protocol version 3. It is used to retrieve email messages from a mail server to a mail client.    | 110                      |
| IMAP     | Internet Message Access Protocol. It is used to access and manipulate email messages on a mail server.        | 143                      |
| DNS      | Domain Name System. It is used to translate domain names into IP addresses.                                   | 53                       |
| DHCP     | Dynamic Host Configuration Protocol. It is used to assign IP addresses and other network parameters to hosts. | 67 (server), 68 (client) |
| SNMP     | Simple Network Management Protocol. It is used to monitor and manage network devices.                         | 161                      |

- A mnemonic to remember some of the application layer protocols and their

port numbers is:

### **Have Fun Sending Mail Postcards In Different Styles**

HTTP (80), FTP (20, 21), SMTP (25), POP3 (110), IMAP (143), DNS (53), DHCP (67, 68), SNMP (161)

### **Domain Name System**

- The Domain Name System (DNS) is a service that translates domain names into IP addresses and vice versa.
- Domain names are human-friendly names that identify websites or other resources on the internet, such as google.com or wikipedia.org.
- IP addresses are numerical identifiers that computers use to communicate with each other over the internet, such as 142.250.74.238 or 208.80.154.224.
- DNS allows users to access websites or resources by typing domain names instead of IP addresses, which are easier to remember and type.
- DNS also allows websites or resources to change their IP addresses without affecting their domain names, which provides flexibility and scalability.
- DNS works as a distributed and hierarchical database that consists of many servers around the world that store and exchange information about domain names and IP addresses.
- The hierarchy of DNS is divided into levels, each with a different scope and authority. The levels are:
  - The root level, which is the highest level and contains information about the top-level domains (TLDs), such as .com, .org, .net, etc.
  - The top-level domains (TLDs), which are the second level and contain information about the second-level domains (SLDs), such as google.com, wikipedia.org, etc.
  - The second-level domains (SLDs), which are the third level and contain information about the subdomains or hosts, such as mail.google.com, en.wikipedia.org, etc.
  - The subdomains or hosts, which are the lowest level and contain information about the specific resources or services, such as www.mail.google.com, en.m.wikipedia.org, etc.
- The process of resolving a domain name into an IP address or vice versa is called DNS lookup or DNS query. It involves the following steps:
  - The user types a domain name into a web browser or an application, such as google.com.
  - The web browser or the application sends a DNS query to a local DNS server, which is usually provided by the user's internet service provider (ISP) or network administrator.
  - The local DNS server checks its cache to see if it has the answer to the query. If it does, it returns the answer to the web browser or the application. If it does not, it forwards the query to a root DNS server.
  - The root DNS server checks its records to see which TLD server is

- responsible for the domain name. It returns the IP address of the TLD server to the local DNS server.
- The local DNS server sends a new query to the TLD server, asking for the IP address of the SLD server that handles the domain name.
- The TLD server returns the IP address of the SLD server to the local DNS server.
- The local DNS server sends another query to the SLD server, asking for the IP address of the host or subdomain that corresponds to the domain name.
- The SLD server returns the IP address of the host or subdomain to the local DNS server.
- The local DNS server returns the final answer to the web browser or the application, which can then connect to the website or the resource using the IP address.
- DNS lookup can be iterative or recursive. In iterative lookup, the local DNS server only asks for a referral to the next server in the hierarchy, and then sends a new query to that server. In recursive lookup, the local DNS server asks the next server in the hierarchy to resolve the query and return the final answer, without sending a new query.
- DNS lookup can also be forward or reverse. In forward lookup, the domain name is translated into an IP address. In reverse lookup, the IP address is translated into a domain name.
- DNS uses two types of records to store and exchange information: resource records (RRs) and name server records (NSs). RRs contain the mapping between domain names and IP addresses, or other information such as mail servers, text records, etc. NSs contain the information about the authoritative servers for each domain or subdomain.
- DNS uses the User Datagram Protocol (UDP) as the default transport protocol for sending and receiving queries and responses. UDP is a connectionless and unreliable protocol that does not guarantee the delivery, order, or integrity of the messages. However, UDP is faster and more efficient than the alternative protocol, the Transmission Control Protocol (TCP), which is a connection-oriented and reliable protocol that guarantees the delivery, order, and integrity of the messages. TCP is used by DNS only when the message size exceeds the UDP limit of 512 bytes, or when the query or response is encrypted

## World Wide Web

- The World Wide Web (WWW) is a collection of different websites around the world, containing different information shared via local servers (or computers) .
- The WWW was invented by Tim Berners-Lee and his colleagues at CERN, an international scientific organization based in Geneva, Switzerland, in 1989 .
- They created a protocol, HyperText Transfer Protocol (HTTP), which

standardized communication between servers and clients .

- HTTP allows users to access web pages by typing a Uniform Resource Locator (URL) in a web browser, such as Google Chrome or Mozilla Firefox.
- A web page is a document that contains text, images, videos, audio, or other multimedia elements, formatted using HyperText Markup Language (HTML).
- HTML uses tags to define the structure and appearance of a web page, such as headings, paragraphs, lists, tables, links, etc.
- A web page can also contain scripts, such as JavaScript, that add interactivity and functionality to the web page, such as validating user input, performing calculations, or displaying dynamic content.
- A web page can also link to other web pages using hyperlinks, which are clickable text or images that direct the user to another URL.
- The WWW is different from the internet, which is the global network of computers and devices that communicate using various protocols, such as TCP/IP, FTP, SMTP, etc.
- The internet is the infrastructure that enables the WWW and other applications, such as email, instant messaging, online gaming, etc.
- The WWW is one of the most popular and widely used applications of the internet, as it allows users to access and share information easily and conveniently.

### **Mnemonics and learning tricks**

- To remember the difference between the WWW and the internet, you can use the following analogy: The internet is like a highway system that connects different cities, while the WWW is like the buildings and attractions that you can visit in each city.
- To remember the meaning of HTTP, you can use the following acronym: HyperText Transfer Protocol = How To Talk Properly.
- To remember the meaning of HTML, you can use the following acronym: HyperText Markup Language = How To Make Layouts.

### **Hyper Text Transfer Protocol**

- Hyper Text Transfer Protocol (HTTP) is a protocol that defines how web browsers and web servers communicate and exchange data over the internet.
- HTTP is based on a request-response model, where a client (browser) sends a request to a server and the server responds with a status code and optionally some data (such as HTML, images, etc.).
- HTTP requests and responses have a similar structure, consisting of a start line, headers, and a body. The start line contains the method, the URL, and the version of HTTP for requests, and the version, the status code, and the reason phrase for responses. The headers provide additional information about the request or response, such as the content type, the

length, the encoding, the cookies, etc. The body contains the actual data being sent or received, such as HTML, JSON, XML, etc.

- HTTP supports several methods, such as GET, POST, PUT, DELETE, etc., that indicate the action to be performed on the resource identified by the URL. For example, GET is used to retrieve a resource, POST is used to create or update a resource, PUT is used to replace a resource, and DELETE is used to delete a resource.
- HTTP uses Uniform Resource Locators (URLs) to identify and locate resources on the web. A URL consists of several components, such as the scheme, the host, the port, the path, the query, and the fragment. For example, in the URL `https://www.example.com:80/index.html?name=Sydney#section1`, the scheme is `https`, the host is `www.example.com`, the port is `80`, the path is `/index.html`, the query is `name=Sydney`, and the fragment is `section1`.
- HTTP is a stateless protocol, meaning that each request and response are independent and do not keep any information about previous interactions. However, HTTP can use cookies to store and send state information between the client and the server. Cookies are small pieces of data that are stored by the browser and sent along with every request to the same server. Cookies can be used for various purposes, such as authentication, personalization, tracking, etc.
- HTTP is also an extensible protocol, meaning that it can be modified and enhanced by adding new methods, headers, status codes, etc. For example, HTTP/1.1 introduced features such as persistent connections, chunked encoding, caching, etc. HTTP/2 introduced features such as binary framing, multiplexing, compression, etc. HTTP/3 introduced features such as using QUIC as the underlying transport layer, etc.

Some mnemonics and learning tricks for HTTP are:

- To remember the structure of HTTP requests and responses, use the acronym SLHB (Start Line, Headers, Body).
- To remember the most common HTTP methods, use the acronym GPPD (GET, POST, PUT, DELETE).
- To remember the most common HTTP status codes, use the acronym 204C5S (200 OK, 404 Not Found, 500 Internal Server Error).

### **Electronic mail in application layer**

- Electronic mail (email) is one of the most widely used and popular applications of the Internet.
- Email allows users to send and receive text messages, images, audio, video and other types of data files to anyone with an email address, regardless of their physical location or time zone.
- Email operates in the application layer of the Internet protocol stack, using other protocols such as SMTP, POP3, IMAP and MIME to perform

different functions.

- SMTP (Simple Mail Transfer Protocol) is the standard protocol for sending email messages from one server to another. SMTP uses a store-and-forward mechanism, which means that the sender's server temporarily stores the email message and then forwards it to the next server in the path to the destination. SMTP uses TCP as the transport layer protocol and port 25 as the default port number.
- POP3 (Post Office Protocol version 3) and IMAP (Internet Message Access Protocol) are two protocols for retrieving email messages from a server. POP3 allows the user to download all the messages from the server to the user's device, and optionally delete them from the server. IMAP allows the user to access and manipulate the messages on the server without downloading them, and also supports multiple folders and synchronization across devices. POP3 uses TCP and port 110, while IMAP uses TCP and port 143.
- MIME (Multipurpose Internet Mail Extensions) is a standard for encoding and formatting non-text data, such as images, audio, video and attachments, in email messages. MIME defines a set of headers and rules for encoding the data in different formats, such as Base64, quoted-printable and multipart. MIME also specifies the content type and disposition of the data, such as image/jpeg, audio/mp3 and attachment.
- A possible mnemonic to remember the four protocols is: **S**end **P**ictures **I**n **M**ail (SMTP, POP3, IMAP, MIME).
- A possible learning trick to understand the difference between POP3 and IMAP is: POP3 is like **P**icking **uP** your mail from the post office and taking it home, while IMAP is like **I**nspecting your mail at the post office and leaving it there.

### **File Transfer Protocol in application layer**

- File Transfer Protocol (FTP) is an application layer protocol that is used to transfer files between local and remote file systems over the Internet .
- FTP runs on top of TCP, which provides reliable and ordered delivery of data packets .
- FTP uses two TCP connections in parallel to transfer a file: a control connection and a data connection .
- The control connection is used to exchange commands and responses between the FTP client and the FTP server . It remains open throughout the entire FTP session .
- The data connection is used to transfer the actual file data between the FTP client and the FTP server . It is opened and closed for each file transfer .

- FTP supports both text and binary files, and can handle different types of file systems and end-of-line characters .
- FTP requires the user to authenticate with a username and password before accessing the server's files . The authentication process is done in plaintext, which means it is not secure and can be intercepted by attackers .
- FTP can operate in two modes: active mode and passive mode . In active mode, the FTP client initiates the data connection to the FTP server. In passive mode, the FTP server initiates the data connection to the FTP client .
- Passive mode is more common and preferred, as it can avoid firewall and NAT issues that may block the data connection in active mode .
- FTP is widely used for uploading and downloading files from web servers, sharing files among users, and backing up data .

### Mnemonics and learning tricks

- To remember the difference between active and passive mode, you can use the following mnemonic: **A**ctive mode = **A**sk for data connection, **P**assive mode = **P**rovide data connection.
- To remember the port numbers used by FTP, you can use the following trick: FTP uses port **21** for the control connection and port **20** for the data connection. The numbers **21** and **20** are in reverse order, just like the letters **F** and **T** in FTP.
- To remember the commands and responses used by FTP, you can use the following table:

| Command | Description                              | Response                                             |
|---------|------------------------------------------|------------------------------------------------------|
| USER    | Sends the username                       | 331: User name okay, need password                   |
| PASS    | Sends the password                       | 230: User logged in                                  |
| CWD     | Changes the working directory            | 250: Requested file action okay                      |
| LIST    | Lists the files in the current directory | 150: File status okay; about to open data connection |
| RETR    | Retrieves a file from the server         | 150: File status okay; about to open data connection |
| STOR    | Stores a file on the server              | 150: File status okay; about to open data connection |
| QUIT    | Terminates the FTP session               | 221: Service closing control connection              |



## Remote login in application layer

- Remote login is a service that allows a user to access a remote system over a network, such as the Internet, and execute commands on that system.
- Remote login is an example of an application layer protocol, which is a set of rules and formats that define how applications communicate with each other and with the lower layers of the network stack.
- Remote login protocols typically use a client-server model, where the client is the user's device and the server is the remote system. The client initiates a connection request to the server, and the server either accepts or rejects it. If accepted, the server authenticates the user and grants access to a shell or a graphical user interface (GUI) on the remote system. The client and the server then exchange data until the connection is terminated by either party.
- Some of the common remote login protocols are:
  - Telnet: Telnet is one of the oldest and simplest remote login protocols. It uses TCP port 23 and sends data in plain text, which makes it insecure and vulnerable to eavesdropping and tampering. Telnet is mainly used for testing and debugging purposes, and not for production environments.
  - Secure Shell (SSH): SSH is a secure and encrypted alternative to Telnet. It uses TCP port 22 and supports various encryption and authentication methods, such as public-key cryptography and password-based login. SSH also allows tunneling of other protocols, such as FTP and X11, over the same connection. SSH is widely used for remote administration and file transfer.
  - Remote Desktop Protocol (RDP): RDP is a protocol that allows a user to access a graphical desktop environment on a remote system. It uses TCP port 3389 and supports various features, such as audio, video, clipboard, printer, and file sharing. RDP is mainly used for remote access to Windows-based systems, but it can also work with other operating systems with compatible software.
  - Virtual Network Computing (VNC): VNC is another protocol that allows a user to access a graphical desktop environment on a remote system. It uses TCP port 5900 and sends pixel data of the remote screen to the client, which renders it on the local screen. VNC is platform-independent and can work with any operating system that supports a graphical display. VNC is less efficient than RDP, but more flexible and customizable.
- Some of the advantages of remote login are:
  - It allows a user to access and control a remote system from anywhere, as long as there is a network connection.

- It reduces the need for physical access and maintenance of the remote system, which can save time, money, and resources.
- It enables a user to perform tasks that require higher processing power, storage capacity, or specialized software on the remote system, which may not be available on the local system.
- It facilitates collaboration and sharing of data and resources among multiple users and systems.
- Some of the disadvantages of remote login are:
  - It depends on the availability and reliability of the network connection, which may be affected by factors such as bandwidth, latency, congestion, and interference.
  - It exposes the remote system to potential security risks, such as unauthorized access, data theft, malware infection, and denial-of-service attacks. Therefore, proper encryption, authentication, and firewall settings are essential for remote login.
  - It may cause performance degradation and resource consumption on both the client and the server, especially if the data exchanged is large or complex.
  - It may create compatibility and interoperability issues among different systems, platforms, and protocols, which may require additional software or configuration.

### **Network management in application layer**

- Network management is the process of configuring, monitoring, and managing the performance and security of a network and its resources, such as routers, switches, servers, printers, etc.
- Network management is usually performed by network administrators using specialized software tools and protocols that communicate with the network devices and collect information about their status, configuration, performance, and faults.
- Network management can be divided into five functional areas: fault management, configuration management, performance management, security management, and accounting management.
- The application layer is the topmost layer of the OSI model, which provides the interface between the network and the user applications, such as web browsers, email clients, file transfer programs, etc.
- The application layer protocols define the rules and formats for exchanging data between different applications over the network, such as HTTP, FTP, SMTP, DNS, etc.
- Some of the application layer protocols are also used for network management purposes, such as Simple Network Management Protocol (SNMP), Common Management Information Protocol (CMIP), and Remote Network Monitoring (RMON).
- SNMP is the most widely used application layer protocol for network man-

agement. It uses UDP port number 161/162 to communicate between network management stations (NMS) and network agents (NA) that are installed on the network devices.

- SNMP allows the NMS to query the NA for information about the device's configuration, performance, and faults, and to receive notifications (traps) from the NA when certain events occur on the device. SNMP also allows the NMS to modify the configuration of the device by sending commands (sets) to the NA.
- SNMP has three versions: SNMPv1, SNMPv2c, and SNMPv3. SNMPv1 is the original and simplest version, but it has limited security and functionality. SNMPv2c adds some enhancements, such as bulk data transfer and 64-bit counters, but it still uses plain text community strings for authentication. SNMPv3 adds security features, such as encryption, authentication, and access control, but it is more complex and requires more resources.
- CMIP is another application layer protocol for network management, which is based on the OSI model and uses TCP port number 163/164. CMIP is more powerful and flexible than SNMP, but it is also more complex and less popular. CMIP supports object-oriented data modeling, event notification, and operation invocation, and it uses Abstract Syntax Notation One (ASN.1) for data encoding.
- RMON is an application layer protocol that defines a set of statistics and functions for remote network monitoring. RMON can be used to monitor traffic, errors, utilization, and performance of network segments and devices. RMON uses SNMP as the transport protocol to communicate between the NMS and the RMON probes that are deployed on the network.
- A mnemonic to remember the five functional areas of network management is FCAPS, which stands for Fault, Configuration, Accounting, Performance, and Security.
- A learning trick to understand the difference between SNMP and CMIP is to compare them with HTTP and SOAP. HTTP is a simple and widely used protocol for web communication, but it has limited security and functionality. SOAP is a more powerful and flexible protocol for web services, but it is more complex and less popular. SOAP uses XML for data encoding, which is similar to ASN.1 used by CMIP.

### **Data compression in application layer**

- Data compression is the function of presentation layer in OSI reference model, but it can also be performed at the application layer in some protocols.
- Data compression allows to reduce the number of bits that need to be transmitted on the network or to optimize disk space when saving data.
- Data compression can be either lossless or lossy. Lossless compression preserves the original data without any loss of information, while lossy compression discards some information to achieve higher compression ra-

tios.

- Some examples of lossless compression algorithms are Huffman coding, Lempel-Ziv-Welch (LZW) coding, and run-length encoding (RLE). Some examples of lossy compression algorithms are JPEG, MPEG, and MP3.
- Data compression can be either static or dynamic. Static compression applies a fixed compression scheme to the data, while dynamic compression adapts the compression scheme to the characteristics of the data.
- Data compression can be either symmetric or asymmetric. Symmetric compression uses the same algorithm for both compression and decompression, while asymmetric compression uses different algorithms for compression and decompression.
- Data compression can be either adaptive or non-adaptive. Adaptive compression changes the compression parameters according to the data, while non-adaptive compression uses fixed parameters for all data.
- Data compression can be either online or offline. Online compression compresses the data as it is transmitted or received, while offline compression compresses the data before or after transmission or reception.
- Data compression can have various advantages and disadvantages depending on the application, such as:
  - Advantages:
    - \* Reduces bandwidth consumption and transmission time
    - \* Saves storage space and cost
    - \* Improves performance and efficiency
    - \* Enhances security and privacy by encrypting the data
  - Disadvantages:
    - \* Increases processing overhead and complexity
    - \* Introduces errors and distortions in lossy compression
    - \* Requires compatibility and standardization of compression algorithms
    - \* May degrade quality of service and user experience
- Some examples of application layer protocols that use data compression are:
  - File Transfer Protocol (FTP): Supports data compression using MODE Z command
  - Simple Mail Transfer Protocol (SMTP): Supports data compression using Content-Encoding header
  - Domain Name System (DNS): Supports data compression using label compression technique
  - Hypertext Transfer Protocol (HTTP): Supports data compression using Accept-Encoding and Content-Encoding headers
  - Secure Shell (SSH): Supports data compression using Compression and CompressionLevel options
- A possible mnemonic to remember the types of data compression is: **LSDS SAON**. It stands for **L**ossless, **L**ossy, **S**tatic, **D**ynamic, **S**ymmetric, **A**symmetric, **A**daptive, **N**on-adaptive, **O**nline, **N**offline.

## Cryptography in application layer

- Cryptography in application layer is the use of cryptographic techniques and protocols to provide security services at the highest layer of the network stack.
- Application layer security can protect the data and messages exchanged by the applications from unauthorized access, modification, or disclosure.
- Application layer security can also provide authentication, integrity, confidentiality, non-repudiation, and access control for the users and applications.
- Some examples of application layer security protocols are:
  - Secure Sockets Layer (SSL) and Transport Layer Security (TLS): These protocols provide secure communication between web browsers and web servers, as well as other applications that use the TCP/IP protocol suite. They use asymmetric cryptography, symmetric cryptography, and digital signatures to establish secure sessions, encrypt and decrypt data, and verify the identity and integrity of the parties.
  - Secure Shell (SSH): This protocol provides secure remote login and command execution between two computers over an insecure network. It uses asymmetric cryptography, symmetric cryptography, and digital signatures to authenticate the users and hosts, encrypt and decrypt the data, and prevent replay attacks.
  - Pretty Good Privacy (PGP) and GNU Privacy Guard (GPG): These protocols provide secure email and file encryption and decryption, as well as digital signatures for message authentication and integrity. They use asymmetric cryptography, symmetric cryptography, and hash functions to encrypt and decrypt the data, and verify the identity and integrity of the senders and receivers.
  - Internet Protocol Security (IPsec): This protocol provides secure communication between two or more computers or networks over the Internet. It uses asymmetric cryptography, symmetric cryptography, and digital signatures to authenticate the peers, encrypt and decrypt the data, and prevent replay attacks. IPsec can operate at the network layer or the transport layer, depending on the mode of operation.
- Some advantages of application layer security are:
  - It can provide end-to-end security, meaning that the data is protected from the source to the destination, without relying on the security of the intermediate nodes or networks.
  - It can be tailored to the specific needs and requirements of the applications and users, such as the level of security, the type of encryption, the key management, etc.

- It can be implemented independently of the underlying network protocols and architectures, making it more flexible and adaptable to different scenarios and environments.
- Some disadvantages of application layer security are:
  - It can introduce additional overhead and complexity, such as the need for encryption and decryption, key generation and distribution, certificate management, etc.
  - It can be vulnerable to attacks that target the application layer, such as malware, phishing, social engineering, etc.
  - It can be difficult to achieve interoperability and compatibility among different applications and platforms that use different security protocols and standards.

### **Basic concepts of cryptography in application layer**

Cryptography is the science of securing communications from adversaries by using mathematical techniques. Cryptography can provide confidentiality, integrity, authenticity, and non-repudiation to the data exchanged between applications. Cryptography can be used in various ways in the application layer, such as:

- Encrypting and decrypting data stored in databases, files, or cloud services.
- Encrypting and decrypting data transmitted over networks, such as web requests, emails, or instant messages.
- Signing and verifying data to prove its origin and integrity, such as digital signatures, certificates, or tokens.
- Generating and managing keys and certificates to enable secure communication and authentication, such as public key infrastructure (PKI), identity-based encryption (IBE), or key exchange protocols.
- Hashing and salting data to protect it from unauthorized modification or disclosure, such as passwords, checksums, or message authentication codes (MACs).

Some of the basic concepts of cryptography that are used in the application layer are:

- **Symmetric key cryptography:** A type of cryptography that uses the same key for both encryption and decryption. Symmetric key cryptography is fast and efficient, but it requires a secure way to distribute the key to the parties involved. Examples of symmetric key algorithms are AES, DES, and RC4.
- **Asymmetric key cryptography:** A type of cryptography that uses a pair of keys for encryption and decryption. One key is public and can be shared with anyone, while the other key is private and must be kept secret. Asymmetric key cryptography is more secure and flexible than symmetric key

cryptography, but it is slower and more complex. Examples of asymmetric key algorithms are RSA, ECC, and DSA.

- Hash function: A function that maps any input to a fixed-length output, called a hash or a digest. Hash functions are one-way, meaning that it is easy to compute the hash from the input, but hard to find the input from the hash. Hash functions are used to verify the integrity and authenticity of data, such as passwords, checksums, or MACs. Examples of hash functions are SHA, MD5, and BLAKE.
- Digital signature: A technique that uses asymmetric key cryptography to sign and verify data. A digital signature is created by encrypting the hash of the data with the private key of the signer, and it can be verified by decrypting the signature with the public key of the signer and comparing it with the hash of the data. Digital signatures are used to prove the origin and integrity of data, such as documents, emails, or certificates. Examples of digital signature algorithms are RSA, DSA, and ECDSA.
- Encryption mode: A technique that specifies how to encrypt and decrypt data using a symmetric key algorithm. Encryption modes are used to enhance the security and efficiency of symmetric key cryptography, by adding randomness, padding, or chaining to the data. Examples of encryption modes are ECB, CBC, CTR, and GCM.